



JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE AND TECHNOLOGY
COLLEGE OF ENGINEERING AND TECHNOLOGY
SCHOOL OF MECHANICAL, MANUFACTURING AND MATERIALS
ENGINEERING
DEPARTMENT OF MARINE ENGINEERING AND MARITIME OPERATIONS
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**DEVELOPMENT AND EVALUATION OF AN AUDITABLE,
RESOURCE-CONSCIOUS PARAMETRIC OPTIMIZATION
WORKFLOW FOR MARINE STRUCTURAL COMPONENTS**

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by

**AMINA AHMED BWANA - ENM241-0087/2022
MBOJE ABDULLAH - ENM241-0088/2022**

Supervised by

SUPERVISOR NAME

*A project report submitted in partial fulfilment of the requirements
for the award of the Degree of Bachelor of Science in Marine Engineering
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DECLARATION

We hereby declare that this Project Report is our original work and has not been presented for a degree in any other university or for any other award.

Name: Amina Ahmed Bwana

REG NUMBER: ENM241-0087/2022

Signature: _____

Date: _____

Name: Mboje Abdullah

REG NUMBER: ENM241-0088/2022

Signature: _____

Date: _____

Supervisor Declaration

This Project Report has been submitted for examination with my approval as the University Supervisor.

Name: Supervisor Name

Signature: _____

Date: _____

Dedication

We dedicate this Project Report to our parents, guardians, lecturers, and mentors. Their encouragement, sacrifice, and guidance supported our academic journey and the completion of this project.

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We thank our supervisor for the technical guidance, critical questions, and academic direction provided throughout this project. We also thank the staff of the Department of Marine Engineering and Maritime Operations for the knowledge and facilities that supported this work. We are grateful to our classmates, friends, and families for their practical help and encouragement. Autodesk educational access and the open-source Python community made it possible to complete the software work without direct project expenditure.

Abstract

This project developed and evaluated the Nelder-Mead Marine Optimizer (NEMO), an auditable workflow for the parameterized mass optimization of marine structural components. The system combines unit-aware engineering definitions, first-order structural screening, seeded Latin-hypercube sampling, bounded multi-start Nelder-Mead search, traceable run records, a file-based Autodesk Fusion bridge, native CAD generation, semantic boundary metadata, validation-package generation, and a results dashboard. Three component archetypes were studied: an Aluminum 6061-T6 equipment bracket, an S275 steel lifting padeye, and an Aluminum 6061-T6 stabilizer fin. Each part used 60 seeded Latin-hypercube samples and three local-search starts. The lightest selected analytically feasible finalists had masses of 0.1134 kg, 3.5478 kg, and 14.3154 kg, which corresponded to reductions of 72.9%, 59.9%, and 27.2% from the respective analytical baselines. Native Fusion CAD mass differed from the analytical estimate by 1.15%, 3.80%, and 0.59% for these finalists. Generator sweeps tested each baseline and 20 seeded design vectors. The bracket and padeye completed all 21 integration cases. The stabilizer generated all 21 artifacts, although the outer test process reached its 30-minute timeout while reading the final response. The bracket also preserved the proof-of-concept path into two exploratory Fusion Static Stress studies. Their images identify the geometry, support and load regions, and qualitative response patterns. The second study, however, used 1.472 N instead of the intended 1472 N, while the first did not provide a complete traceable load definition. Their numerical results were therefore rejected. The work shows that a resource-conscious, human-reviewed workflow can reduce routine search and CAD effort while keeping analytical, CAD, and FEA claims separate. The reported designs remain screening candidates. They require corrected, mesh-converged FEA, design-code checks, verified loads and materials, fatigue and fabrication assessment, and qualified engineering review before manufacture or service.

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LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviations

API	Application Programming Interface
CAD	Computer-Aided Design
CAE	Computer-Aided Engineering
CFD	Computational Fluid Dynamics
CLI	Command-Line Interface
FEA	Finite Element Analysis
FOS	Factor of Safety
IPC	Inter-Process Communication
JSON	JavaScript Object Notation
JKUAT	Jomo Kenyatta University of Agriculture and Technology
LHS	Latin-Hypercube Sampling
NACA	National Advisory Committee for Aeronautics
NEMO	Nelder-Mead Marine Optimizer
POC	Proof of Concept
SI	International System of Units
STEP	Standard for the Exchange of Product Model Data
STL	Stereolithography mesh format
V&V	Verification and Validation

Symbols

A	Area (m^2)
b	Stabilizer span (m)
C_L	Lift coefficient
E	Young's modulus (Pa)
F_d	Design load (N)
$F_{\min}$	Minimum permitted factor of safety
g	Acceleration due to gravity (m/s^2)
I	Second moment of area (m^4)
$J(\mathbf{x})$	Penalized optimization objective
K_t	Stress-concentration factor
m	Mass (kg)
M	Bending moment (N m)
ρ	Density (kg/m^3)
σ	Normal or equivalent stress (MPa)

σ_y	Material yield strength (MPa)
τ	Shear stress (MPa)
V	Volume (m ³) or speed (m/s), as identified in context
w	Penalty weight
$\mathbf{x}$	Physical design vector
$\mathbf{z}$	Scaled design vector
δ	Displacement (mm)
$\delta_{\max}$	Maximum permitted displacement (mm)

Chapter 1

Introduction

This chapter establishes the engineering context, defines the problem addressed by the study, and presents its aim, objectives, significance, and limits. The three component case studies are introduced together because they provide different tests of the same optimization and CAD–analysis workflow.

1.1 Background

Marine vessels and offshore systems contain many secondary structural components, including machinery brackets, lifting points, appendage roots, stiffeners, doublers, and attachment details. Although small relative to the supporting vessel or platform, these components transfer concentrated or distributed loads into a larger structure. Their behaviour can therefore affect equipment alignment, lifting safety, vibration, watertight integrity, and the durability of adjacent structure.

The load paths are often three-dimensional. A bracket can combine baseplate and rib bending, fastener reactions, bearing, contact, and local stress at fillets or welds. A padeye can combine pin bearing, net-section tension, shear-out, lug-root stress, gusset action, and weld load transfer. A stabilizer fin can combine distributed pressure, root bending, shear, torsion, shell action, local buckling, fatigue, and hydrodynamic effects. A credible design study must therefore distinguish the behaviour represented by its models from the failure modes that require later investigation.

Mass reduction is useful because structural mass affects material use, vessel displacement, payload, installation effort, fuel demand, and centre-of-gravity management. It cannot, however, be pursued independently of strength, stiffness, stability, fatigue life, corrosion allowance, manufacturability, inspection, and damage tolerance. The design task is therefore formulated as constrained mass minimization.

Traditional development commonly alternates between preliminary calculations, manual CAD changes, and selected finite element analyses. This approach can produce acceptable designs, but it examines only a small portion of a multi-variable design space

and can separate dimensions, loads, material data, boundary conditions, and results into unrelated records. Parametric modelling offers a more systematic basis for iteration. A family of geometries can be evaluated, compared, and reduced to a small set of candidates for higher-fidelity work. Research has connected commercial CAD and CAE tools to automated optimization loops [1]; however, geometric change also alters the faces used for loads, supports, contacts, and mesh controls. The engineering meaning of these regions must be preserved, an issue described as the preservation of simulation intent [2].

The Nelder-Mead Marine Optimizer (NEMO) addresses this problem through a tiered workflow. First-order analytical models guide design-space exploration and optimization. Autodesk Fusion then rebuilds native geometry and provides physical-property evidence for selected candidates. Final FEA remains a human-reviewed activity because the public Fusion Python API used in this project did not provide a dependable route to initiate the required Static Stress solves and extract every required result. The workflow consequently keeps analytical predictions, CAD properties, and solver evidence as distinct claims.

Three components were selected to test the approach. Their principal characteristics are summarized in Table 1.1.

Table 1.1: Case-study components and represented structural behaviour

Part	Material and load	Parameterized features	Principal behaviour represented
Bracket	Aluminum 6061-T6; 1.4715 kN equipment envelope	Baseplate, ribs, and fillet	Concentrated equipment load transferred through a ribbed plate to four mounting holes
Padeye	S275 steel; 14.715 kN pin-load envelope	Doubler, lug, neck, gussets, and fillet	Pinned lifting load transferred through a lug and reinforcement into a supporting plate
Stabilizer	Aluminum 6061-T6; 11.808 kN prescribed resultant	Skin, spars, ribs, insert, and root fillet	Distributed transverse load carried by a fixed-envelope internal structure to a root flange

The bracket provides the complete proof of concept, from parameter definition and analytical search to native CAD and a manually prepared Static Stress study. The padeye and stabilizer demonstrate that the architecture can represent different variables, materials, loads, geometry, and boundary roles. Their manual FEA campaigns remain

future validation work.

1.2 Problem Statement

Manual design iteration does not systematically explore the combined effects of plate dimensions, stiffener geometry, fillet radius, lug proportions, spar position, and reinforcement size. Conversely, solving a detailed FEA model for every candidate would require more time, computation, and software interaction than a resource-constrained student project can support. A practical method is needed to use inexpensive models for broad exploration and reserve manual FEA for a small, traceable candidate set.

The method must also preserve evidence integrity. The engineering definition, design variables, calculated metrics, generated geometry, load and support regions, and validation record must remain connected. A generated solid is not a successful analysis if a required boundary region is absent; a stale CAD response must not be associated with a new design; CAD mass must not be presented as structural validation; and a completed solver report must be rejected when its applied load is incorrect.

The resulting research gap is not a lack of optimization algorithms or CAD software. It is the need for a compact, inspectable workflow that combines bounded design-space exploration, failure-contained evaluation, native CAD generation, semantic boundary identification, and an explicit separation of analytical, CAD, and FEA evidence on an ordinary engineering workstation.

1.3 Aim and Objectives

The project aims to develop and evaluate an auditable, resource-conscious workflow for the parameterized mass optimization and validation preparation of marine structural components.

The specific objectives are to:

1. define unit-aware parameter spaces, materials, loads, constraints, analytical screening models, and boundary roles for a bracket, padeye, and stabilizer;
2. explore each bounded design space using seeded Latin-hypercube sampling and scaled, multi-start Nelder–Mead optimization while retaining invalid evaluations as controlled records;
3. generate native Fusion geometry, physical properties, exchange artifacts, and semantic boundary metadata through a versioned CAD bridge;
4. verify each CAD generator at its baseline and 20 seeded design vectors, and prepare traceable baseline and finalist validation sets;

5. evaluate the lower-mass analytical candidates and compare analytical mass with native-CAD mass; and
6. demonstrate the bracket proof of concept and identify the further evidence required before any manufacturing or service decision.

These objectives are examined through four questions:

1. Can one part-aware workflow represent and process the three component types without embedding bracket-specific assumptions in shared operations?
2. Can seeded sampling and bounded multi-start search identify lower-mass, analytically feasible candidates for all three parts?
3. Can native CAD and the required load and support regions remain usable across bounded geometry changes, and how closely does CAD mass agree with the analytical estimates?
4. What does the bracket proof of concept reveal about the evidence required before the candidates can be accepted as structural designs?

1.4 Significance of the Study

The study demonstrates how low-cost analytical screening can reduce the number of candidates requiring manual CAD and FEA attention while retaining the information needed to review each decision. Its contribution is an implementation-and-evaluation workflow rather than a new optimization algorithm.

The engineering definitions keep variables, units, bounds, materials, loads, constraints, and required boundary roles consistent across evaluation, optimization, candidate selection, and CAD generation. Scaled coordinates, bounded trials, explicit feasibility filtering, and controlled failed evaluations prevent one invalid design from ending a search or being promoted as an acceptable finalist. The CAD handoff records physical properties and identifies engineering regions such as fixed supports, equipment-load faces, pin-bearing surfaces, pressure bands, and displacement monitors. These records do not replace engineering judgement, but they make the intended load path inspectable.

The workflow is also suitable for an educational setting because assumptions and evidence levels remain visible. It operates using existing student hardware, educational software access, open-source Python packages, and local data exchange. These resources have economic value, but the study required no new direct project expenditure.

1.5 Scope and Limitations

The implemented study covers three registered component definitions, first-order linear-elastic screening, six bracket variables, nine padeye variables, nine stabilizer variables, seeded sampling, bounded multi-start optimization, recorded run results, native Fusion generation, STEP and optional STL export, semantic boundary metadata, CAD sweeps, and baseline and finalist validation preparation. A results dashboard supports review of the recorded optimization data but does not provide a separate calculation model.

The analytical models estimate mass, a governing stress, yield-based factor of safety, and displacement. The bracket uses an equivalent cantilever representation; the padeye compares net-section, bearing, shear-out, and lateral-bending estimates; and the stabilizer uses root-section bending and shear under a prescribed distributed load. The models assume homogeneous, isotropic, linear-elastic material and declared design envelopes. They do not resolve detailed contact, fastener preload, weld behaviour, three-dimensional local stress, nonlinear response, shell buckling, fatigue, corrosion, impact, vibration, fluid–structure interaction, or hydrodynamic pressure prediction.

The project uses software, analytical, CAD, and FEA evidence as separate levels. Software tests and analytical results are complete for the reported scope, and the CAD campaigns provide geometry, mass, artifacts, and boundary-role evidence. The bracket includes two exploratory solver studies, but neither supplies acceptable structural results: the second study records 1.472 N rather than the intended 1472 N, while the first does not provide a complete traceable load definition. Padeye and stabilizer FEA remain future work.

The finite sample and three local starts do not establish a global optimum. The defensible optimization claim is the best feasible design found within the declared parameterized search. The project also does not establish classification compliance, fabrication readiness, or service safety. Acceptance would require verified loads and materials, applicable design-code checks, calibrated and mesh-converged FEA, relevant fatigue, buckling, corrosion, contact, fastener, and weld assessments, physical testing where required, and qualified engineering review.

The values in this section are first-order screening estimates. They are not certified engineering results and do not establish manufacturing or service suitability.

Chapter 2 reviews the relevant optimization, CAD–CAE, and structural-analysis literature. Chapter 3 presents the models and study method. Chapter 4 reports and discusses the analytical, CAD, and bracket-study evidence. Chapter 7 states the conclusions, achievement of the objectives, limitations, recommendations, and future work.

Chapter 2

Literature Review

This chapter reviews the established methods on which NEMO is based. The review covers constrained structural optimization, derivative-free search, Nelder–Mead, variable scaling, Latin-hypercube sampling, tiered model fidelity, CAD–CAE integration, simulation intent, marine structural applications, and computational-model credibility. It then relates these topics to the bracket, padeye, and stabilizer case studies and defines the limited research gap addressed by this project.

2.1 Structural Optimization Methods

2.1.1 Constrained Search

Structural optimization formalizes the selection of design variables to improve an objective while satisfying constraints. An objective can represent mass, cost, compliance, displacement, stress, fatigue damage, environmental impact, or a weighted combination of measures. Constraints can represent strength, stiffness, stability, vibration, geometry, fabrication, inspection, or regulatory limits. The problem is written in the general form

$$\underset{\mathbf{x}}{\text{minimize}} \quad f(\mathbf{x}), \tag{2.1}$$

$$\text{subject to} \quad g_j(\mathbf{x}) \leq 0, \quad j = 1, \dots, m, \tag{2.2}$$

$$h_k(\mathbf{x}) = 0, \quad k = 1, \dots, p, \tag{2.3}$$

$$\mathbf{l} \leq \mathbf{x} \leq \mathbf{u}, \tag{2.4}$$

where $\mathbf{x}$ is the design vector, f is the objective, g_j and h_k are inequality and equality constraints, and $\mathbf{l}$ and $\mathbf{u}$ are variable bounds.

Structural optimization is commonly divided into size, shape, and topology optimization. Size optimization changes dimensions such as thickness and section area. Shape optimization changes geometric boundaries while retaining a general feature ar-

range. Topology optimization changes the distribution or connectivity of material inside a design region. These classes can overlap, but the distinction helps define what an optimization claim means.

NEMO performs size and bounded parametric shape optimization. The number and general arrangement of major features remain fixed. The optimizer changes dimensions such as plate thickness, rib height, lug width, spar position, and fillet radius. It does not remove arbitrary material or create a new topology. This restriction keeps every candidate compatible with an intended load path, a native CAD generator, and predefined load and support roles.

The project uses mass as the principal objective. This choice is easy to measure and compare, but it does not make mass the only design criterion. A low-mass candidate is useful only if its constraints pass and its omitted failure modes are later checked. In marine practice, this qualification is especially important because cyclic loading, corrosion, fabrication detail, and uncertain operational loads can govern a design that passes one static yield check.

2.1.2 Derivative-Free Optimization

Many engineering analyses appear as black-box functions to the optimizer. The optimizer supplies dimensions and receives calculated outputs, but it does not have exact derivatives of those outputs. CAD regeneration, meshing changes, solver tolerances, discrete feature behavior, and failed candidates can also make numerical gradients unreliable.

Direct-search and derivative-free methods use objective values without requiring analytical gradients. Kolda, Lewis, and Torczon reviewed the development of direct-search methods and distinguished methods with formal convergence properties from more heuristic searches [3]. Their review shows that derivative-free operation is not the same as assumption-free operation. The designer must still scale variables, handle bounds, define constraints, select termination criteria, and evaluate sensitivity to the starting point.

Derivative-free methods can be attractive when each function evaluation is moderately expensive and the number of variables is limited. They are less attractive when the dimension is very high or when gradients can be calculated cheaply and reliably. The three NEMO cases contain six or nine continuous variables. Their analytical evaluations are inexpensive, while the wider workflow includes potential discontinuities from geometry checks. This combination supports the use of a simple derivative-free method for the reference implementation.

2.1.3 The Nelder–Mead Simplex Method

Nelder and Mead introduced the simplex method in 1965 [4]. In an n -dimensional space, the method maintains $n + 1$ vertices. It evaluates and ranks these vertices, calculates the centroid of the better vertices, and attempts to replace the worst vertex. The standard

trial operations are reflection, expansion, contraction, and shrink.

If $\mathbf{x}_h$ is the worst vertex and $\bar{\mathbf{x}}$ is the centroid of the remaining vertices, a reflected point is

$$\mathbf{x}_r = \bar{\mathbf{x}} + \alpha(\bar{\mathbf{x}} - \mathbf{x}_h), \quad (2.5)$$

where α is the reflection coefficient. A sufficiently good reflection can lead to expansion:

$$\mathbf{x}_e = \bar{\mathbf{x}} + \gamma(\mathbf{x}_r - \bar{\mathbf{x}}). \quad (2.6)$$

An unsuccessful reflection can cause contraction toward the centroid. When contraction also fails, the simplex shrinks toward its best vertex. The standard coefficients used by NEMO are $\alpha = 1$, $\gamma = 2$, $\rho = 0.5$, and $\sigma = 0.5$.

The method has several practical advantages. It is compact, does not require a gradient, and is easy to explain and implement. It also has important limits. Lagarias *et al.* established convergence results under restricted conditions and discussed cases where the method can approach a non-minimizer [5]. Nelder–Mead is a local search. Its final design can depend on the initial simplex, variable scaling, bounds, penalties, and the shape of the objective landscape.

These limitations affect the language of this report. One local run cannot support a claim of global optimality. NEMO uses a seeded design-space sample and three starts to reduce dependence on the configured baseline. The report uses the phrase “best feasible design found within this parameterized search.” This wording is more defensible than “global optimum.”

Table 2.1: Nelder–Mead operations and their purpose in the implemented search

Operation	Coefficient	Purpose
Reflection	$\alpha = 1$	Tests a point across the centroid from the current worst vertex.
Expansion	$\gamma = 2$	Extends a successful reflected step when it improves the best objective.
Contraction	$\rho = 0.5$	Tests a shorter step when reflection does not give adequate improvement.
Shrink	$\sigma = 0.5$	Moves all non-best vertices toward the best point after failed trial steps.

2.1.4 Variable Scaling, Bounds, and Constraint Penalties

Engineering variables can differ greatly in magnitude and unit. A plate thickness may be 4 mm, a chordwise spar position may be 30%, and a root insert length may be 180 mm. If an optimizer works directly in these physical coordinates, its simplex geometry and trial step sizes can be dominated by the numerically largest variables.

NEMO maps each physical variable x_i to the scaled coordinate

$$z_i = \frac{x_i - l_i}{u_i - l_i}, \quad 0 \leq z_i \leq 1, \quad (2.7)$$

where l_i and u_i are the lower and upper bounds. The evaluator performs the inverse mapping before it uses the engineering equations. Trial points are clipped at the scaled bounds.

Scaling gives each coordinate the same numerical range. It does not make the physics equally sensitive. A small thickness change can alter both mass and section properties strongly, while a similar scaled change in a fillet radius may affect only one stress factor in the analytical model. Sensitivity remains a property of the model.

Nelder–Mead does not directly enforce general nonlinear constraints in this implementation. NEMO therefore uses a penalty objective. If $m(\mathbf{x})$ is mass, $F(\mathbf{x})$ is factor of safety, $F_{\min}$ is the minimum factor of safety, $\delta(\mathbf{x})$ is displacement, and $\delta_{\max}$ is the displacement limit, the objective is

$$J(\mathbf{x}) = m(\mathbf{x}) + w \left[\left(\frac{\max(0, F_{\min} - F(\mathbf{x}))}{F_{\min}} \right)^2 + \left(\frac{\max(0, \delta(\mathbf{x}) - \delta_{\max})}{\delta_{\max}} \right)^2 \right], \quad (2.8)$$

with $w = 100$ for the current parts.

A feasible design has $J = m$. An infeasible design has a higher objective according to its normalized violations. Penalties simplify the search, but their weight affects ranking. A weak penalty can make an infeasible light design more attractive than a feasible heavier design. A large penalty can create a steep landscape near the boundary. The final selection must therefore filter explicitly for feasibility. The stabilizer case in Chapter 4 demonstrates why this check is necessary.

2.1.5 Latin-Hypercube Sampling

McKay, Beckman, and Conover introduced Latin-hypercube sampling as a stratified method for computer experiments [6]. For each variable, an N -point Latin hypercube samples once from each of N equal-probability intervals. Independent permutations combine the one-dimensional samples into N multi-dimensional points.

Compared with an unstructured random sample of the same size, a Latin hypercube improves one-dimensional stratification. It does not guarantee dense coverage of every interaction, especially in high dimensions. Sixty points remain sparse in a nine-dimensional space. The method is therefore best viewed here as an economical exploration step.

NEMO uses 60 samples per part with seed 42. The fixed seed makes the campaign repeatable. Sampling serves three purposes:

1. it checks whether calculated outputs change in a physically sensible direction over the bounds;
2. it identifies low-mass feasible candidates that can start local search; and
3. it reduces reliance on one baseline-start Nelder–Mead run.

Sampling also provides diagnostic evidence. A high failure rate can reveal poor parameter bounds or a fragile geometry generator. An unexpectedly high feasibility rate can reveal an over-conservative baseline or a screening model that does not activate the declared constraints.

2.2 Model Fidelity and CAD–CAE Integration

2.2.1 Tiered Model Fidelity

Engineering optimization often combines models with different cost and accuracy. Perherstorfer, Willcox, and Gunzburger review multi-fidelity methods that combine inexpensive low-fidelity models with expensive high-fidelity models [7]. Formal multi-fidelity optimization uses the high-fidelity model inside the optimization strategy to improve prediction, selection, or convergence.

NEMO does not yet implement such a closed loop. The analytical model guides the search. Native CAD verifies geometry and mass for selected designs. Manual FEA follows as a separate validation stage. The correct description is a *tiered-fidelity workflow*. The analytical optimum is not automatically corrected by FEA.

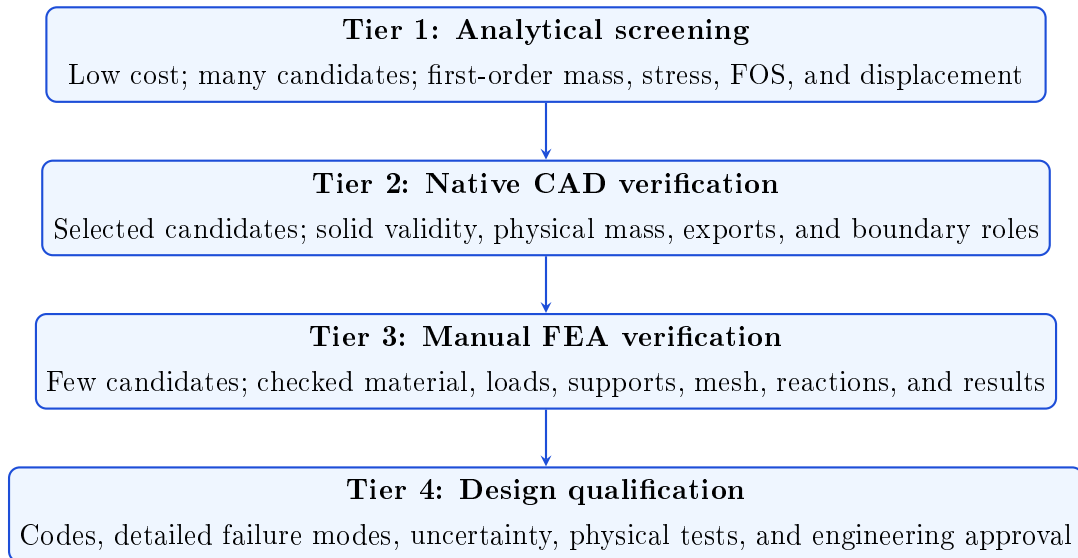


Figure 2.1: Tiered evidence model used by the project

The distinction between the tiers protects the report from a common error: treating agreement on one quantity as complete validation. Close agreement between analytical and CAD mass shows that the volume approximations represent the generated geometry reasonably well. It does not show that the analytical stress model represents local three-dimensional stress.

2.2.2 CAD–CAE Integration

Automated structural optimization requires dependable communication between geometry, analysis, and optimization. Park and Dang coupled commercial CAD and CAE software through scripting, programming interfaces, and metamodels [1]. Their work shows how integration can reduce repetitive work and analysis cost. It also introduces an additional source of error when a surrogate model approximates the high-fidelity response.

Li *et al.* developed a unified parametric representation based on extended voxels. Their method supports a closed loop among design, simulation, and optimization while carrying design and simulation meaning [8]. This work confirms that automated CAD–CAE optimization is an established research area. NEMO does not claim novelty from automation alone.

The project instead uses a modest file-based architecture. External Python handles engineering definitions, optimization, run records, and validation packages. Fusion handles native CAD and physical properties. Versioned JSON files connect the two environments. This approach is slower than an in-process call but has useful properties:

- a request and response can be inspected with an ordinary text editor;
- the Fusion environment does not need third-party optimization packages;
- a failed request can preserve a controlled error record;

- the protocol can retain compatibility with the original bracket proof of concept; and
- the same external application can later support another CAD or FEA backend.

The architecture also has limits. A single shared request directory serializes CAD work. Polling adds latency. Local file locking requires retry behavior. The bridge cannot make an unavailable solver API available. These are accepted trade-offs for an inspectable student-scale implementation.

2.2.3 Simulation Intent and Semantic Boundary Roles

Geometry changes can invalidate an analysis even when the regenerated solid looks correct. A load attached to a low-level face identifier can disappear or move after a feature changes the boundary representation. A reliable parametric workflow must preserve why a region exists and how it is used.

Nolan *et al.* define simulation intent as the high-level modeling and idealization decisions required for fit-for-purpose analysis [2]. Their work associates analysis meaning with regions and interfaces rather than relying only on unstable topology.

NEMO applies the same principle in a smaller implementation. The bracket declares `fixed_support` and `equipment_load`. The padeye declares `fixed_support` and `pin_bearing`. The stabilizer declares `fixed_support`, four pressure bands, and `tip_monitor`. The Fusion generator selects faces with part-specific geometric rules and records area, centroid, normal, bounding box, and cylinder data where available.

Semantic roles improve traceability, but they do not remove the need for human inspection. A broad geometric selector can match an unintended face. A topology change can divide one intended region into several faces. The manual FEA procedure must display and confirm every selected face before solving.

2.3 Marine Structural Applications

Marine structural optimization must represent realistic loads, supports, and failure modes. Gentils, Wang, and Kolios optimized an offshore wind support structure with parametric FEA and a genetic algorithm [9]. Their constraints included stress, displacement, vibration, buckling, and fatigue. The study reported a 19.8% mass reduction for the complete support structure.

Zheng, Li, and Xiao later used sensitivity analysis, surrogate models, and a genetic algorithm for jacket structures [10]. Their method addressed natural frequency, displacement, stress, buckling, and later fatigue checks. Wang, Mantey, and Zhang developed a fast parametric FEA and genetic-algorithm tool for jacket structures [11]. These studies show that marine structural optimization commonly requires several load cases and

constraints beyond static yield.

The literature provides two lessons for NEMO. First, mass reduction must remain subordinate to the governing engineering constraints. Second, the selected design can change when supports and loads change. The project therefore treats its load cases as declared engineering envelopes, not as complete vessel or offshore design cases.

Table 2.2: Relationship between selected literature and the NEMO implementation

Source area	Established lesson	Use in NEMO
Direct search [3–5]	Derivative-free local search is useful but needs scaling, stopping rules, and careful claims.	Scaled bounded variables, three starts, and “best design found” wording.
Computer experiments [6]	Stratified samples improve economical exploration.	Sixty seeded Latin-hypercube samples per part.
Multi-fidelity methods [7]	Models of different cost require a defined relationship.	Tiered analytical, CAD, manual FEA, and qualification evidence.
CAD–CAE integration [1,8]	Automation must preserve data and analysis meaning.	Versioned JSON bridge, native CAD, and explicit partial responses.
Simulation intent [2]	Boundary meaning must survive geometry change.	Semantic roles and geometric face signatures.
Marine optimization [9–11]	Mass is one objective among strength, stiffness, stability, dynamics, and fatigue.	Static screening followed by a stated list of required detailed checks.

2.3.1 Bracket Structural Behavior

Autodesk lists brackets among components suited to linear Static Stress analysis when the assumptions of small deformation, unchanged load direction, and linear elastic material behavior hold [12]. The apparent simplicity of a bracket can be misleading. Local stress depends on load introduction, fastener restraint, contact, rib geometry, fillet radius, plate flexibility, and weld detail.

The NEMO bracket is an equipment-foundation archetype with a baseplate, four mounting holes, two ribs, and rib-root fillets. Its analytical model uses an equivalent cantilever relation. This model captures the broad effects of section depth, plate thickness, rib thickness, and load eccentricity. It does not resolve bolt bearing, fastener preload, prying, weld-toe stress, local contact, or three-dimensional restraint.

The bracket is useful as a proof of concept because the geometry is easy to recognize and the load path is easy to explain. It is not useful because it is free from modeling risk.

The proof-of-concept FEA report later showed that a three-order-of-magnitude input error can survive through a completed solver run. This finding supports a workflow that checks input values and reaction balance before it discusses result contours.

2.3.2 Padeyes and Pinned Connections

Padeyes transfer lifting force through a pin hole into a lug, reinforcement, welds, and supporting structure. Potential governing effects include pin-bearing stress, net-section tension, shear-out, bending, local contact, lug-root stress, gusset load transfer, and weld stress. Clearance, load angle, out-of-plane load, and local geometry can strongly affect the stress field.

Soh and Soh compared simplified two-dimensional padeye models with a three-dimensional finite-element solution [13]. Their work showed that model selection affects stress prediction and can lead to conservative overdesign. Liu, Zhou, and Tan used FEA to study padeye stress and deformation and then formulated an optimization problem to reduce steel use [14]. These studies support the use of a simplified model for early screening, provided later FEA represents the important three-dimensional behavior.

Pedersen examined the stress concentration around a pin-loaded hole and showed the influence of contact modeling and local geometric refinement [15]. This result is directly relevant to the NEMO padeye. Its current analytical model compares several nominal stress modes, but it does not model a separate pin, clearance, nonlinear contact, or a weld group. A later FEA campaign must decide whether a distributed bearing load is adequate for screening or whether explicit pin contact is required.

Marine lifting appliances also require a governing rule set. DNV-ST-N001 addresses marine operations and associated structural considerations [16]. Lloyd's Register publishes a Code for Lifting Appliances in a Marine Environment that covers design, materials, fabrication, testing, marking, survey, and documentation [17]. NEMO does not claim compliance with either source. They are cited to show the wider qualification work that lies beyond a single yield-based optimization.

2.3.3 Stabilizer Fins and Prescribed Hydrodynamic Loads

The stabilizer differs from the bracket and padeye because its load is distributed over an external lifting surface. The project retains a NACA 0015 section. Abbott, von Doenhoff, and Stivers documented NACA airfoil families and aerodynamic data in NACA Report 824 [18]. The use of a known profile makes the geometric definition reproducible, but it does not supply a complete marine pressure field.

The NEMO stabilizer keeps its span, root chord, tip chord, sweep, trailing edge, root flange, and rib stations fixed. It changes skin thickness, two spar positions, two spar thicknesses, rib thickness, root-insert dimensions, and root fillet radius. Its prescribed resultant is calculated from a simplified lift expression and distributed through

four spanwise weights.

This setup supports internal structural sizing under one declared load envelope. It does not optimize hydrodynamic efficiency. It omits vessel motions, unsteady flow, free-surface effects, ventilation, cavitation, actuator behavior, pressure reversal, impact, and fluid–structure interaction. A later design stage must derive the pressure from an accepted hydrodynamic method, CFD, model testing, or full-scale data.

2.4 Finite Element Verification and Validation

2.4.1 Linear Static Analysis

Autodesk describes a Static Stress study as appropriate when deformation is small, load direction does not change significantly, material properties are constant, and the response remains linear elastic [12]. A complete setup requires geometry, materials, constraints, loads, contacts, mesh, solution, and result review [19].

These requirements are coupled. A correct material with a wrong load gives a wrong result. A correct load on the wrong face can change the load path. An overly rigid support can reduce deformation and create a local stress singularity. A coarse mesh can underpredict a local stress. A maximum contour value can occur at a mathematical singularity that is unsuitable for direct acceptance.

Reaction forces provide a basic equilibrium check. Autodesk notes that support reactions should oppose the total applied load [20]. For the bracket, a correctly configured 1472 N downward load should produce a vertical support reaction close to 1472 N, subject to sign convention and small numerical differences. A reaction near 1.472 N would immediately expose the factor-of-one-thousand load error.

Mesh convergence is also necessary. Autodesk explains that coarse meshes commonly underpredict stress and that stress is more mesh-sensitive than displacement [21]. Comparison should use the same geometric location across refinements. A credible record should report the global mesh, local refinements, element order, node and element count, the monitored quantity, and the change between successive meshes.

2.4.2 Verification, Validation, and Model Credibility

Verification asks whether equations and software were solved or implemented correctly. Validation asks whether the model represents the real system well enough for its intended use. These questions are related but distinct. A correctly implemented simplified model can still be unsuitable for a service decision.

ASME V&V 10 provides a framework and terminology for verification, validation, and uncertainty quantification in computational solid mechanics [22]. NASA-STD-7009B emphasizes intended use, acceptance criteria, input pedigree, uncertainty, sensitivity,

verification, validation, and communicated credibility [23]. The principles are relevant to NEMO even though the project does not fall under NASA governance. Model credibility depends on the decision being supported.

Table 2.3: Evidence levels and the questions they answer

Evidence level	Primary question	Representative checks
Software verification	Did the code implement its declared contracts and calculations consistently?	Unit tests, bounds, baseline feasibility, failure handling, logging schema, repeatable seeds.
CAD verification	Did the requested dimensions create a usable native solid and required artifacts?	One connected solid, positive volume, STEP export, boundary roles, physical mass.
FEA verification	Was the numerical study configured and solved correctly?	Material, load, support, contact, reaction balance, mesh convergence, result location.
Physical validation	Does the model represent measured component behavior adequately for its intended use?	Proof load, strain, displacement, failure mode, repeatability, uncertainty.
Design qualification	Does the complete design satisfy governing service requirements?	Design codes, load combinations, fatigue, buckling, corrosion, fabrication, inspection, competent approval.

The table shows why a project can complete software and CAD verification without completing structural validation. It also explains why a single FEA contour is not final proof. The bracket report is retained because it documents the setup and exposes a process weakness. Its incorrect numerical result is not promoted to a higher evidence level.

2.5 Literature Synthesis and Research Gap

The literature establishes every main method used by NEMO. Nelder–Mead provides a compact derivative-free local search [4, 5]. Direct-search literature explains the need for careful implementation and claims [3]. Latin-hypercube sampling provides a repeatable exploration stage [6]. Multi-fidelity literature explains the value and limits of combining models of different cost [7]. CAD–CAE research identifies integration and simulation intent as central problems [1, 2, 8]. Marine optimization studies show that useful lightweighting depends on several constraints and credible load models [9–11]. Computational-mechanics guidance requires a declared intended use and an evidence

chain [22, 23].

No single method in NEMO is new. The reviewed literature also contains more advanced automated loops, global optimizers, surrogate models, and high-fidelity marine applications. The project therefore does not claim to be the first marine optimizer or the first CAD–CAE automation system.

The limited gap addressed here is practical integration at student-project scale. The reviewed sources do not directly provide a transparent, part-manifest-driven workflow that combines unit-aware definitions, first-order screening, seeded sampling, scaled bounded multi-start search, failure-contained evaluation, versioned atomic file exchange, native Fusion CAD, semantic boundary metadata, validation packages, and explicit separation of analytical, CAD, and FEA evidence across a bracket, padeye, and stabilizer.

NEMO addresses that implementation gap as an auditable reference workflow. Its value lies in exposing each assumption and handoff, reducing repetitive candidate work, and preserving enough evidence for a supervisor or engineer to check the result. The next chapter describes how the workflow was implemented and evaluated.

Chapter 3

Methodology

This chapter describes the research design, engineering models, design-space exploration, optimization method, native CAD generation, and validation approach. The bracket is used as the complete proof of concept, while the padeye and stabilizer demonstrate the extension of the method to different structural forms and load paths.

3.1 Research Design

The project used a design-and-evaluate research method. It created an engineering software artifact, tested the artifact, applied it to three case studies, and evaluated the resulting analytical and CAD evidence against declared requirements. The work did not use experimental human subjects or a statistical population. Its evidence came from software behavior, analytical calculations, native CAD artifacts, and a manually configured solver report.

The work proceeded through four technical phases:

1. **Engineering definition:** define parameters, bounds, units, baselines, materials, loads, constraints, fixed geometry, and boundary roles.
2. **Search:** evaluate baselines, sample the design spaces, and run three bounded Nelder–Mead starts for each part.
3. **CAD verification:** rebuild selected designs in Fusion, calculate physical properties, export artifacts, and check boundary roles.
4. **Validation preparation:** package baselines and finalists, prepare manual Static Stress studies, and record the evidence required for engineering acceptance.

The phases follow the evidence tiers in Figure 2.1. A later tier can reject an earlier candidate. For example, CAD failure can reject an analytically feasible design. Corrected FEA can reject a design that passes both analytical and CAD checks.

3.1.1 Workflow Architecture

The workflow contains four functional layers: a shared engineering definition, the analytical search, the Fusion CAD handoff, and the resulting evidence record. Figure 3.1 shows how these layers support the study.

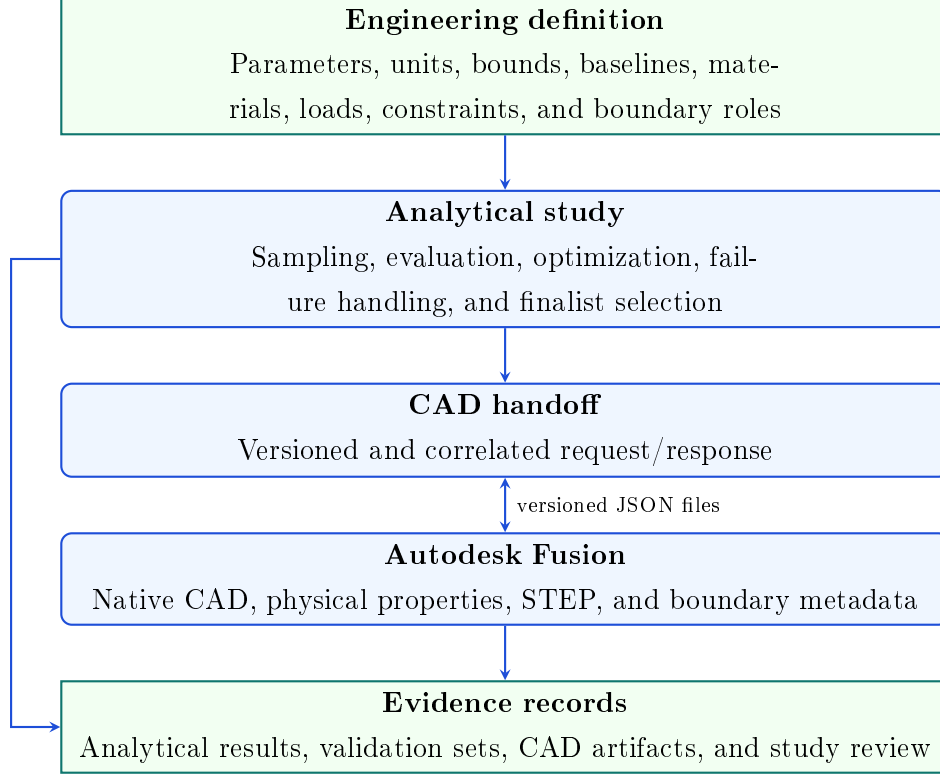


Figure 3.1: NEMO study workflow and evidence flow

The analytical and CAD stages run in separate Python environments. Their separation allows the search to use the required scientific dependencies while Fusion retains its native application environment. A small versioned exchange passes candidate parameters to Fusion and returns geometry properties and boundary evidence. Autodesk defines the application lifecycle for Fusion scripts and add-ins [24].

3.2 Engineering Models

3.2.1 Common Engineering Definition

Each component was represented by one engineering definition containing its design variables, units, lower and upper bounds, baseline dimensions, material properties, design load, acceptance criteria, fixed geometry, and required boundary roles. Most dimensions are expressed in millimetres; the stabilizer spar locations are percentages of local chord. Lengths were converted to metres for the SI calculations, while stress and displacement were reported in MPa and millimetres. The same definition governed analytical evalu-

ation, optimization, candidate selection, and CAD generation so that a change in load, bound, or unit was not applied differently at successive stages.

3.2.2 Bracket

The bracket represents a foundation for a 50 kg auxiliary unit. The design load uses a project dynamic factor of 3:

$$F_d = mg\gamma_d = 50(9.81)(3) = 1471.5 \text{ N.} \quad (3.1)$$

The dynamic factor is an engineering envelope. It is not a modal, shock, or vessel-motion result. The analytical calculation retains 1471.5 N. The current Fusion input dialog does not accept the decimal force value used in this workflow. The manual study therefore uses the conservative integer value 1472 N, a difference of approximately 0.034%.

The project-defined Aluminum 6061-T6 properties are density 2700 kg/m³, yield strength 276 MPa, Young's modulus 68.9 GPa, and Poisson ratio 0.33. Table 3.1 gives the variables.

Table 3.1: Bracket design variables

Variable	Unit	Lower	Baseline	Upper
Baseplate length	mm	100	150	200
Baseplate width	mm	80	100	150
Baseplate thickness	mm	4	8	15
Rib height	mm	20	40	60
Rib thickness	mm	3	6	10
Fillet radius	mm	2	5	10

The fixed geometry includes four 10 mm mounting holes and two ribs. The screening acceptance criteria are $FOS \geq 2.5$ and $\delta \leq 0.5$ mm. These limits are project criteria and do not replace a governing marine rule set.

Analytical formulation

All geometric dimensions enter the equations in meters. The approximate volume contains the rectangular baseplate, four-hole subtraction, two triangular ribs, and a fillet contribution:

$$V_b = LWt_b - n_h\pi \left(\frac{d_h}{2}\right)^2 t_b + n_r \frac{Lh_r t_r}{2} + n_r \frac{\pi r_f^2}{4} \max(0.35W, t_r). \quad (3.2)$$

The bracket mass is

$$m_b = \rho V_b. \quad (3.3)$$

The equivalent cantilever moment is

$$M_b = F_d e, \quad e = \max(0.03, 0.35L + 0.02). \quad (3.4)$$

The effective section modulus combines the baseplate and rib terms:

$$Z_{\text{eff}} = \frac{W t_b^2}{6} + 0.65 n_r \frac{t_r h_r^2}{6}. \quad (3.5)$$

The stress estimate includes a fillet-dependent concentration factor:

$$K_t = \max \left(1.08, 1.35 - \frac{r_f [\text{mm}]}{40} \right), \quad (3.6)$$

$$\sigma_b = \frac{M_b}{Z_{\text{eff}}} K_t. \quad (3.7)$$

The effective second moment of area is

$$I_{\text{eff}} = \frac{W t_b^3}{12} + 0.45 n_r \frac{t_r h_r^3}{36}, \quad (3.8)$$

and the tip-equivalent displacement is

$$\delta_b = \frac{F_d e^3}{3 E I_{\text{eff}}}. \quad (3.9)$$

The yield-based factor of safety is

$$FOS_b = \frac{\sigma_y}{\sigma_b}. \quad (3.10)$$

The equations preserve the expected broad trends. Increased plate or rib thickness increases mass and stiffness. Increased rib height strongly increases bending stiffness. Increased fillet radius reduces the assumed stress concentration. The model does not calculate bolt bearing, bolt preload, weld-group stress, weld-toe fatigue, local plate bending, contact, or three-dimensional load sharing.

3.2.3 Padeye

The padeye represents a lifting point for a 500 kg working mass with a dynamic factor of 3:

$$F_d = 500(9.81)(3) = 14\,715 \text{ N}. \quad (3.11)$$

The project-defined S275 properties are density 7850 kg/m³, yield strength

275 MPa, Young’s modulus 210 GPa, and Poisson ratio 0.30. The fixed pin-hole diameter is 50 mm, and two gussets reinforce the lug. The criteria are $FOS \geq 2.5$ and $\delta \leq 0.5$ mm.

Table 3.2: Padeye design variables

Variable	Unit	Lower	Baseline	Upper
Base length	mm	220	260	320
Base width	mm	150	180	230
Base thickness	mm	8	14	22
Lug height	mm	140	180	230
Lug thickness	mm	12	20	30
Neck width	mm	80	100	140
Gusset height	mm	70	110	120
Gusset thickness	mm	6	10	18
Fillet radius	mm	8	15	25

Analytical formulation

The volume model combines the doubler plate, a tapered lug and circular crown, the 50 mm hole subtraction, two triangular gussets, and a fillet approximation. This representation follows the native generator closely enough for a mass consistency check, but it is not an exact boundary-representation calculation.

The structural model calculates four candidate stresses. The net-section tension stress uses the remaining ligament at the hole and a fillet-dependent stress factor. The nominal bearing stress is

$$\sigma_{\text{bearing}} = \frac{F_d}{d_p t_l}, \quad (3.12)$$

where d_p is pin diameter and t_l is lug thickness. The shear-out estimate uses the two load paths between the hole and the outer edge:

$$\tau_{\text{out}} = \frac{F_d}{2A_{\text{shear}}}. \quad (3.13)$$

A lateral-bending estimate represents an assumed 25% side-load component. The reported padeye stress is

$$\sigma_p = \max \left(\sigma_{\text{net}}, \sigma_{\text{bearing}}, \sqrt{3} \tau_{\text{out}}, \sigma_{\text{lateral}} \right). \quad (3.14)$$

The displacement combines axial and lateral terms, and the factor of safety is $FOS_p = \sigma_y / \sigma_p$. This model ranks candidates by several relevant nominal modes. It does not model a separate pin, pin clearance, nonlinear contact, local peak stress, weld

geometry, or fatigue. These limitations define the later FEA and design-code work.

3.2.4 Stabilizer

The stabilizer retains a NACA 0015 outer envelope with 800 mm span, 500 mm root chord, 250 mm tip chord, 12-degree sweep, and 8 mm trailing edge. It has a 560 mm by 160 mm by 18 mm root flange and three ribs at 25%, 50%, and 75% span.

The project-defined Aluminum 6061-T6 properties match the bracket. The prescribed resultant follows

$$F_d = \frac{1}{2} \rho_w V^2 S C_L \gamma_d, \quad (3.15)$$

where seawater density is 1025 kg/m³, speed is 8 m/s, planform area is 0.3 m², lift coefficient is 0.8, and the dynamic factor is 1.5. Thus

$$F_d = \frac{1}{2} (1025) (8^2) (0.3) (0.8) (1.5) = 11\,808 \text{ N}. \quad (3.16)$$

Four pressure bands use relative weights 1.00, 0.93, 0.75, and 0.40. These are declared engineering assumptions, not CFD data. The screening criteria are $FOS \geq 2.5$ and tip displacement no more than 5 mm.

Table 3.3: Stabilizer design variables

Variable	Unit	Lower	Baseline	Upper
Skin thickness	mm	4	7	12
Front spar position	% chord	25	30	38
Rear spar position	% chord	58	65	72
Front spar thickness	mm	4	8	14
Rear spar thickness	mm	4	6	12
Rib thickness	mm	3	5	9
Root insert length	mm	120	180	260
Root insert thickness	mm	8	12	20
Root fillet radius	mm	12	25	40

Analytical formulation

The volume model combines the two-sided skin, front and rear spar webs, three internal ribs, root insert, fixed flange, and root-fillet approximation. The NACA thickness equation estimates the local spar height at each chord position. The root section combines skin, spar, and insert contributions to the second moment of area.

The root bending moment uses a resultant centroid at 40% span:

$$M_r = 0.40F_db. \quad (3.17)$$

The model calculates a nominal root bending stress σ_b and shear stress τ . A root-fillet factor K_t modifies the bending term. The equivalent stress is

$$\sigma_s = \sqrt{(K_t\sigma_b)^2 + 3\tau^2}. \quad (3.18)$$

The tip displacement estimate is

$$\delta_s = \frac{F_db^3}{8EI_{\text{eff}}}. \quad (3.19)$$

The model does not derive the pressure field. It also omits torsional deflection, shell buckling, panel crippling, local skin-spar stress, fatigue, fluid-structure interaction, and root-bearing detail. It therefore supports structural screening under a prescribed load, not hydrodynamic optimization.

3.3 Design-Space Exploration and Optimization

3.3.1 Evaluation and Failure Handling

Each evaluator returns a structured record with volume, mass, maximum stress, factor of safety, maximum displacement, objective, status, and error. It first checks that the part exists, all required parameters are present, values are finite, values lie within bounds, and the requested evaluation mode is available.

Invalid inputs and unavailable evaluators return a failed record with objective 10^9 . The analytical optimizer can continue after such a record. A CAD-only response has partial status because it can contain valid mass and artifacts while structural metrics remain unavailable. The reserved `open_fea` mode deliberately returns a controlled failure. No Gmsh or CalculiX result is fabricated.

3.3.2 Latin-Hypercube Exploration

Each part used 60 Latin-hypercube samples with seed 42. The sampler operated in the scaled unit cube. Each variable was divided into 60 intervals. A random point was selected within each interval, and independently permuted variable columns formed the multi-dimensional vectors. The inverse scaling in Equation 2.7 produced physical values.

The sampler evaluated all points, retained failed records, and counted feasible candidates. A feasible candidate met both the factor-of-safety and displacement constraints. Promising feasible sampled designs supplied two of the three optimization starts.

3.3.3 Bounded Multi-Start Optimization

The pure-Python optimizer operated on scaled coordinates. Each part used three starts:

1. the configured baseline;
2. a promising low-mass feasible sample; and
3. a second promising feasible sample that provided another basin or parameter combination.

Each run permitted at most 80 simplex iterations. The optimizer converted each trial point to physical units, evaluated the part, cached repeated vectors, and appended each unique evaluation to the run log. The simplex stopped at the iteration limit or when the objective spread met the configured tolerance.

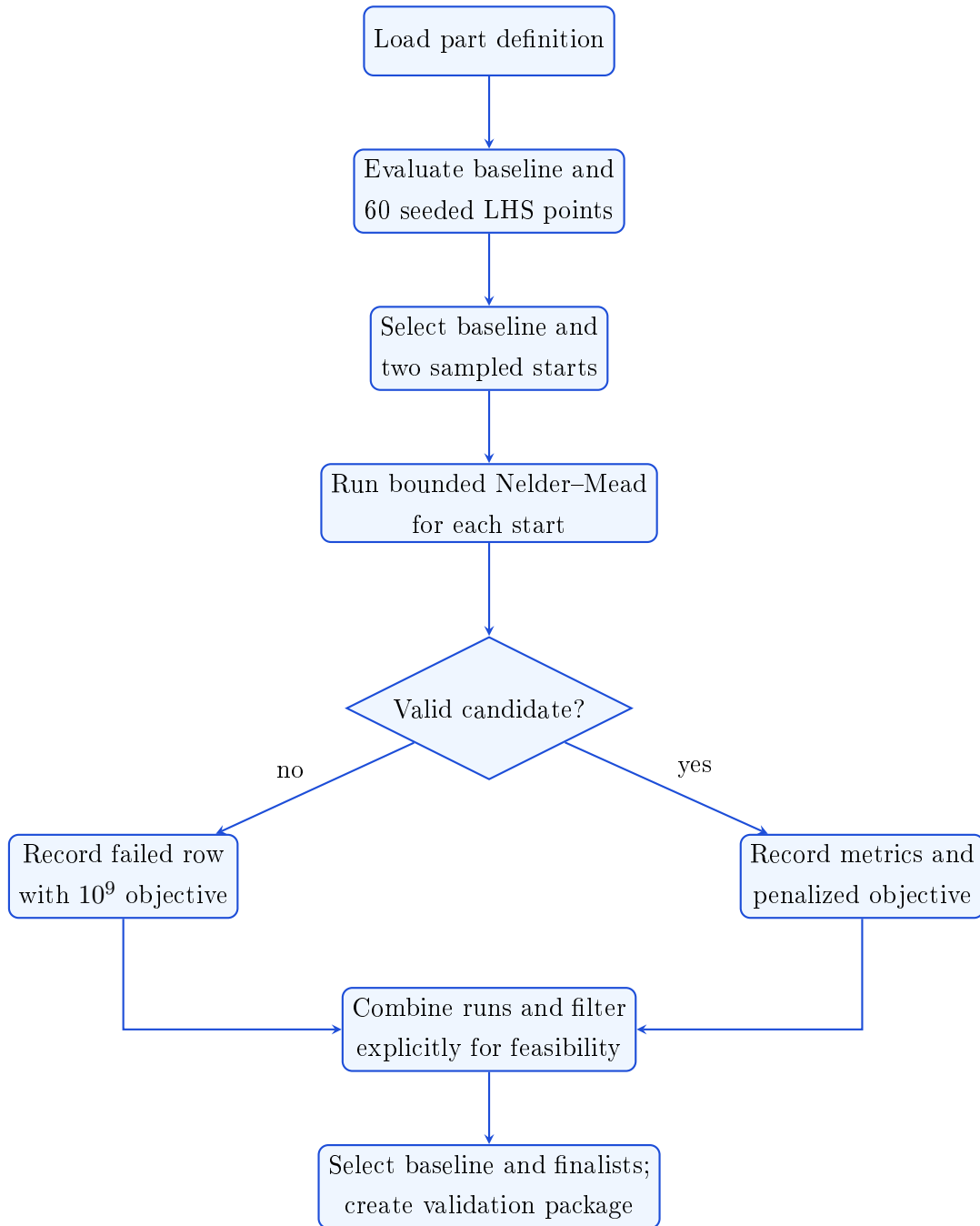


Figure 3.2: Sampling, optimization, and finalist-selection procedure

The selection step is deliberately separate from the optimizer’s minimum objective. The validation package sorts only feasible rows by mass. It retains five finalists so that the lightest design does not become the only candidate for later FEA.

3.3.4 Results Recording and Review

Each evaluated candidate was recorded with its part identifier, parameter values, calculated metrics, feasibility status, objective, and any evaluation error. Run-level metadata preserved the seed and search context. A dashboard was used to inspect convergence, parameter activity, mass, constraint use, and failed evaluations without changing the

underlying results.

3.4 CAD Generation and Boundary Definition

3.4.1 Versioned CAD Handoff

Each CAD request identifies the schema, component, run, iteration, evaluation mode, parameter values, and required artifacts. A legacy form is retained for the original bracket demonstration. Requests and responses are written atomically, and each returned result is accepted only when its run and iteration identifiers match the candidate under evaluation. This prevents a partial or stale exchange from being interpreted as current evidence.

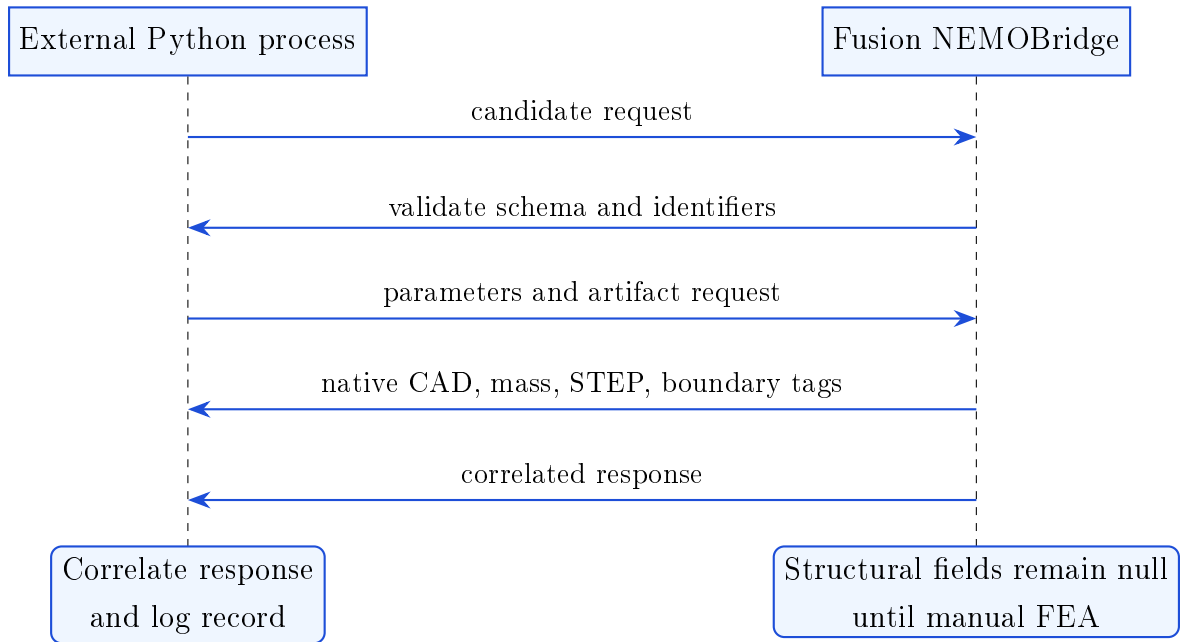


Figure 3.3: Serialized Fusion request and response sequence

The handoff permits one active CAD request at a time. This serialized approach favours traceability over throughput and prevents two candidates from competing for the same exchange state.

3.4.2 Native CAD Generation

The Fusion generator uses the same engineering definitions as the analytical study while remaining within Fusion’s separate Python environment.

Bracket

The bracket generator creates a rectangular baseplate, cuts four mounting holes, creates two triangular ribs, joins the bodies, and applies four long rib-root fillets. It checks that the result is one connected solid with positive volume. It identifies the mounting-hole

cylindrical faces as the fixed-support role and the four upper sloping rib faces as the equipment-load role.

Padeye

The padeye generator creates the doubler plate, tapered lug, circular crown, pin bore, two root fillets, and two overlapping gussets. It requires one connected solid. The underside of the doubler plate supplies the fixed-support role. The cylindrical pin-hole region supplies the pin-bearing role.

Stabilizer

The stabilizer generator creates the root flange, lofted NACA outer envelope, root fillet, hollow interior, front and rear spars, three ribs, root insert, and pressure-band splits. It requires one connected solid. The root flange provides the fixed support. Four surface regions provide the pressure bands, and a tip region provides the displacement monitor.

Autodesk exposes physical-property and export functions through the public API [25, 26]. NEMO records mass and volume from the generated component and exports STEP for downstream exchange. STL is optional because it is a tessellated artifact rather than the preferred solid exchange format.

3.4.3 Semantic Boundary Definition

For each required role, the generator stores one or more face signatures. Available fields include area, centroid, normal, bounding box, and cylindrical geometry. The generator fails the sweep acceptance check if a required role has no matching face.

This metadata has three functions. First, it documents what the generator believed the load or support region to be. Second, it helps the engineer find the same region in the manual study. Third, it makes topology problems visible during a sweep.

It does not automatically apply the FEA boundary condition. The engineer must still display the faces, compare them with the intended load path, and reject an incorrect selection.

3.5 Verification and Validation

3.5.1 CAD Generator Verification

Each generator was tested with its configured baseline and 20 seeded Latin-hypercube vectors. The acceptance criteria for each of the 21 cases were:

1. the request completed without an uncontrolled exception;
2. the result contained exactly one connected solid;

3. the physical volume was positive;
4. STEP export completed;
5. boundary metadata export completed; and
6. every required semantic boundary role matched at least one face.

The sweep does not solve FEA. It tests whether the generator and boundary contract remain usable across the bounded design space. It is intentionally separate from the offline test suite because it requires an open Fusion session and mutates the shared live channel.

3.5.2 Finalist Validation Preparation

For each component, the baseline and five light feasible candidates were assembled into a validation set. Each record retained its parameters, analytical predictions, intended material and boundaries, CAD status and mass, and fields for later FEA inputs, outputs, convergence evidence, and engineering disposition. This preserved the connection between a solver image and the candidate to which it belonged.

3.5.3 Bracket Static Stress Studies

The bracket proof of concept continued from the analytical finalist set into Fusion Static Stress. Aluminum 6061 was assigned to the generated bracket, the four mounting-hole surfaces represented the initial fixed-support idealization, and the four upper sloping rib faces represented the equipment load interface. The intended total downward load was 1472 N, obtained by rounding the analytical design load of 1471.5 N to the integer input used for the study.

Two exploratory studies were reviewed. Their material assignment, boundary regions, mesh information, result magnitudes, and result locations were compared with the engineering definition. The review also checked whether the applied force was a total load rather than a load repeated on every selected face. The images and findings from both studies are presented and interpreted in Section 4.4.

3.5.4 Static Stress Assessment Criteria

Study credibility was assessed from the material properties, applied-load magnitude and direction, support and contact idealizations, mesh order and refinement, reaction balance, result locations, and stability under mesh change. Maximum von Mises stress, minimum factor of safety, and maximum displacement were considered only after these input and numerical checks. Peaks at constrained edges were treated cautiously because they can represent local singular behaviour. Even an acceptable linear-static study would cover

only the stated load case and would not establish fatigue, buckling, weld, bolt, corrosion, impact, or classification compliance.

3.5.5 Extension to Padeye FEA

The padeye study will follow the same evidence sequence with part-specific boundaries and result regions. The underside of the doubler plate will provide the first fixed-support idealization. The 14 715 N design load will act through the pin-bearing region.

An initial linear screening model can use a distributed bearing load on the hole surface. A later detailed model should include a separate pin, clearance, contact, and any required nonlinear behavior. The review regions are the loaded hole, crown, net section, shear-out ligaments, lug root, gusset roots, doubler interface, and weld lines. Reaction balance must close to 14 715 N in the applied direction.

The final design assessment must also check the selected lifting code, load angle range, dynamic factors, material certification, weld class, fabrication tolerance, inspection, proof load, fatigue, and supporting structure.

3.5.6 Extension to Stabilizer FEA

The stabilizer study will fix the root-flange interface. The 11 808 N resultant will be distributed over four pressure bands. The band forces are calculated by normalizing the weights 1.00, 0.93, 0.75, and 0.40. Pressure on each band equals its force divided by its actual CAD face area.

The study will monitor tip displacement and inspect the root fillet, front and rear spars, skin-spar intersections, rib junctions, insert termination, root flange, and trailing edge. Shell or solid element choice must represent the thin skins and local joints adequately. Buckling and fatigue require separate studies.

The prescribed load distribution is an interim structural envelope. A later hydrodynamic stage must define operational conditions and derive pressure from an accepted method. The structural model should then include positive and negative load cases, torsion, actuator loads, and relevant vessel motions.

3.5.7 Reproducibility and Method Limitations

Software verification covered engineering-definition loading, bounds, baseline feasibility, qualitative model sensitivity, sampling, optimizer behaviour, controlled failures, results recording, CAD handoff, and finalist preparation. Fusion sweeps were conducted separately because they depend on an active desktop application; they checked geometry and boundary evidence but did not solve FEA.

Exact finalist parameters, analytical results, CAD masses, and solver-study evidence were retained so that the tables and figures in this report could be traced to the

corresponding candidate. The method improves reproducibility, but it does not remove uncertainty in the analytical assumptions, candidate selection, CAD boundary identification, or manual solver setup.

The values in this section are first-order screening estimates. They are not certified engineering results and do not establish manufacturing or service suitability.

The method was designed to reduce candidate count and improve traceability. It was not designed to certify marine hardware. A future engineering decision must use verified loads, applicable codes, detailed FEA, uncertainty assessment, fatigue and fabrication checks, physical testing where required, and qualified approval.

Chapter 4

Results and Discussion

This chapter reports software verification, analytical baselines, Latin-hypercube exploration, multi-start optimization, finalist selection, native CAD sweeps, CAD mass checks, and the bracket FEA quality-control finding. Analytical, CAD, and FEA results are reported separately. The chapter then answers the research questions and discusses the limitations of the evidence.

4.1 Verification Results

The offline verification campaign passed 27 tests; three Fusion-dependent tests were excluded from that campaign because they require an active desktop session. The completed checks covered the engineering definitions and bounds, baseline feasibility, expected analytical trends, repeatable sampling, bounded optimization, failed-evaluation containment, results recording, CAD handoff, candidate preparation, and results review. Native-CAD sweeps were evaluated separately in Section 4.3.1; they did not perform FEA.

4.2 Analytical Optimization Results

4.2.1 Baselines

The values in this section are first-order screening estimates. They are not certified engineering results and do not establish manufacturing or service suitability.

Table 4.1 gives the configured baselines. All three passed the implemented factor-of-safety and displacement criteria.

Table 4.1: Analytical baseline screening results

Part	Mass (kg)	Stress (MPa)	FOS	Deflection (mm)	Volume (m ³)
Bracket	0.418125	41.5321	6.6455	0.195642	0.000154861
Padeye	8.847625	46.1070	5.9644	0.013429	0.001127086
Stabilizer	19.655593	86.9853	3.1730	3.336750	0.007279849

The bracket and padeye baselines had large yield-based margins. Their deflection estimates were also well below the project limits. The stabilizer had less factor-of-safety and displacement margin, although it remained analytically feasible. These differences affected the available mass reduction. A highly conservative baseline provides more room for a mass-only optimizer than a baseline already close to an active constraint.

4.2.2 Latin-Hypercube Exploration

Each part used 60 samples with seed 42. Table 4.2 summarizes the feasible counts and lightest feasible sampled designs.

Table 4.2: Seeded Latin-hypercube results

Part	Feasible samples	Lightest mass (kg)	FOS	Deflection (mm)
Bracket	55 of 60	0.245687	10.4945	0.073331
Padeye	60 of 60	5.923561	4.9324	0.010449
Stabilizer	47 of 60	16.482802	2.7674	4.107580

The sample stage achieved its three intended purposes. It exercised broad combinations of bounded variables, identified feasible low-mass starting points, and showed that the parts had different constraint activity.

The bracket had five infeasible sampled designs, but its lightest feasible sample still had high factor-of-safety margin. This indicates that broad random combinations did not automatically exploit the dimensions that control the screening stress.

All padeye samples were feasible. The result indicates that the declared bounds and simplified stress model produce a generous feasible region under the single project load case. It does not prove that every bounded padeye would pass pin contact, weld, fatigue, or governing lifting rules.

The stabilizer had 13 infeasible samples. Its lightest feasible sample was already close to both constraints compared with the other parts. The stabilizer therefore presented the clearest constrained search.

4.2.3 Multi-Start Optimization

Table 4.3 gives the lightest feasible row found in each local run. The start names identify the configured baseline and two sample-derived starts.

Table 4.3: Lightest feasible result from each Nelder–Mead start

Part	Start	Evaluations	Mass (kg)	FOS	Deflection (mm)
Bracket	Baseline	132	0.126545	2.5530	0.475840
	Sample 1	126	0.116928	2.5208	0.396650
	Sample 2	129	0.113396	2.5386	0.369832
Padeye	Baseline	108	3.837083	2.8528	0.026150
	Sample 1	116	3.609310	2.5767	0.026789
	Sample 2	109	3.547772	2.6224	0.027933
Stabilizer	Baseline	110	14.414066	2.5041	3.932243
	Sample 1	116	14.315361	2.5149	3.822300
	Sample 2	111	14.391670	2.5331	3.847508

Every part benefited from more than one start. The lightest bracket and padeye feasible designs came from the second sampled start. The lightest stabilizer feasible design came from the first sampled start. If the study had used only the configured baseline, the selected masses would have been higher for all three parts.

The bracket baseline-start result approached the 0.5 mm displacement limit, while the other two bracket results were governed more clearly by the factor-of-safety limit. The padeye displacement remained far from 0.5 mm in all runs, so its yield-based stress controlled the search. The stabilizer operated close to the factor-of-safety limit while using approximately 76–79% of the permitted displacement.

4.2.4 Selected Finalists

The validation packages selected the lightest explicitly feasible candidates. Table 4.4 compares the selected finalist with its baseline.

Table 4.4: Selected analytical finalist compared with baseline

Part	Baseline (kg)	Finalist (kg)	Reduction	Final FOS	Defl. (mm)
Bracket	0.418125	0.113396	72.9%	2.5386	0.369832
Padeye	8.847625	3.547772	59.9%	2.6224	0.027933
Stabilizer	19.655593	14.315361	27.2%	2.5149	3.822300

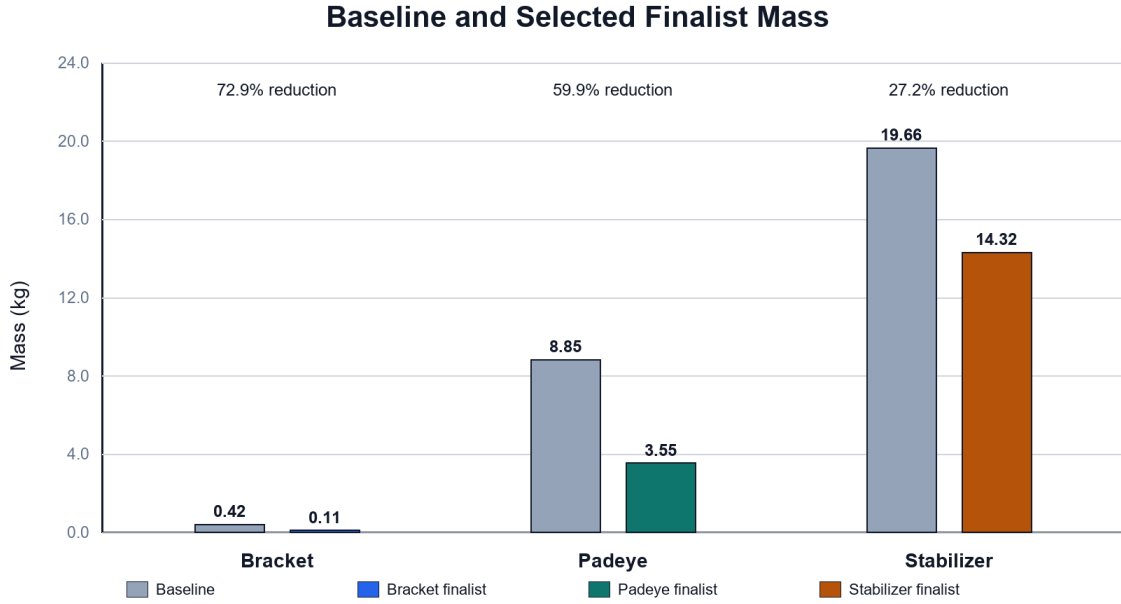


Figure 4.1: Analytical baseline and selected-finalist mass. Percentages are screening reductions within the declared parameterized searches.

The reductions are large, particularly for the bracket and padeye. They must be interpreted with the baseline and model scope. The baselines were configured reference designs, not independently weight-optimized industrial designs. The equations omit several failure modes. The percentage reduction therefore measures the opportunity found by the implemented screening model, not a validated service weight saving.

4.2.5 Finalist Constraint Use

Figure 4.2 normalizes the finalist responses. A factor-of-safety ratio of 1.0 means the finalist is exactly at the minimum factor of safety. A displacement ratio of 1.0 means it is exactly at the displacement limit.

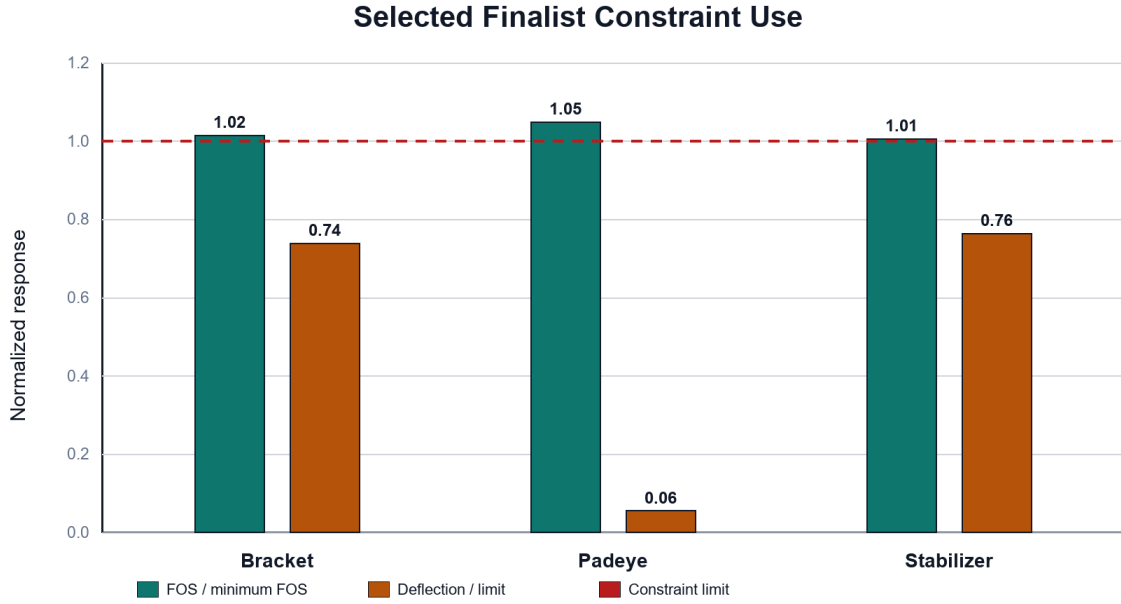


Figure 4.2: Normalized constraint use for the selected analytical finalists

All three finalists have factor-of-safety ratios only slightly above 1.0. This result is expected from a mass-only search with a yield constraint: the optimizer removes material until a constraint becomes active. The result also shows why screening finalists need conservative backup candidates. A small change in load, material, stress concentration, or model bias could make the lightest row infeasible.

The bracket displacement ratio is approximately 0.74. The padeye displacement ratio is approximately 0.056, so the simplified padeye search is almost entirely controlled by stress. The stabilizer displacement ratio is approximately 0.76. A later FEA model can change which constraint governs.

4.2.6 Parameter Activity

Bracket

The selected bracket used approximately 100.03 mm baseplate length, 80.94 mm width, 4.04 mm baseplate thickness, 33.26 mm rib height, 3.00 mm rib thickness, and 3.53 mm fillet radius. The length, width, plate thickness, and rib thickness were near their lower bounds.

The pattern reflects the analytical equations. Plan dimensions and thickness reduce volume directly. Rib height retains section depth and bending resistance with less mass than a uniformly thick plate. This is a plausible broad trend, but the candidate has little allowance for corrosion, fabrication tolerance, weld size, bearing, or local plate action. The lower-bound activity is therefore a prompt to review the bounds and omitted constraints, not proof that the minimum gauge is suitable.

Padeye

The selected padeye used approximately 220.39 mm base length, 150.47 mm base width, 8.05 mm base thickness, 141.73 mm lug height, 12.16 mm lug thickness, 80.00 mm neck width, 70.00 mm gusset height, 6.21 mm gusset thickness, and 25.00 mm fillet radius.

Many thickness and envelope variables approached their lower bounds, while the fillet radius reached its upper bound. In the analytical model, a large fillet reduces the assumed concentration factor with a relatively modest volume cost. The optimizer therefore spends material on the fillet while reducing other dimensions. A three-dimensional contact and weld model may change this balance.

Stabilizer

The selected stabilizer used approximately 4.01 mm skin, 25.0% front-spar position, 72.0% rear-spar position, 7.85 mm front spar, 6.53 mm rear spar, 5.70 mm ribs, 221.48 mm root insert, 13.66 mm insert thickness, and 18.33 mm root fillet.

The skin approached its lower bound, while the spars moved apart. A larger spar separation can increase effective section depth. The optimizer retained root reinforcement to protect the factor-of-safety limit. This response is consistent with the root-section model. It remains sensitive to the assumed pressure distribution and omitted torsion and buckling.

4.2.7 Penalty Objective and Explicit Feasibility

The smallest penalized objective in each stabilizer run was infeasible:

Table 4.5: Infeasible minimum-objective rows from the stabilizer searches

Start	Objective	Mass (kg)	FOS
Baseline	13.9702	13.74695	2.38188
Sample 1	13.7477	13.61934	2.41042
Sample 2	13.7335	13.61934	2.41554

The penalty increased the objective, but the low mass remained attractive enough to produce a smaller value than the feasible rows. If the package had selected the minimum objective without a feasibility filter, it would have promoted a failed design. The implemented selection avoided this error.

This result shows a limitation and a strength. The fixed penalty weight is not strong enough to make every run's objective minimum feasible. However, the system does not treat the penalty as the acceptance rule. Explicit feasibility is the acceptance rule. Future work can compare adaptive penalties, feasibility-first ranking, or a constrained derivative-free method.

4.3 CAD Verification Results

4.3.1 Native CAD Generator Sweeps

The final sweeps used generator version 2.4. Table 4.6 reports their status.

Table 4.6: Baseline plus 20-vector native CAD sweep results

Part	Artifacts created	Integration passes	Recorded observation
Bracket	21 of 21	21 of 21	Completed in approximately 238.19 s.
Padeye	21 of 21	21 of 21	Completed in approximately 157.19 s.
Stabilizer	21 of 21	Final read timed out	All 21 artifacts were created. The outer test reached its 30-minute timeout while reading the final response; direct artifact audit confirmed generation.

Every audited case contained one connected solid, positive volume, STEP output, boundary metadata, and non-empty required boundary roles. The result supports generator behavior over the seeded bounded vectors. It does not prove behavior for every possible point in the continuous design space.

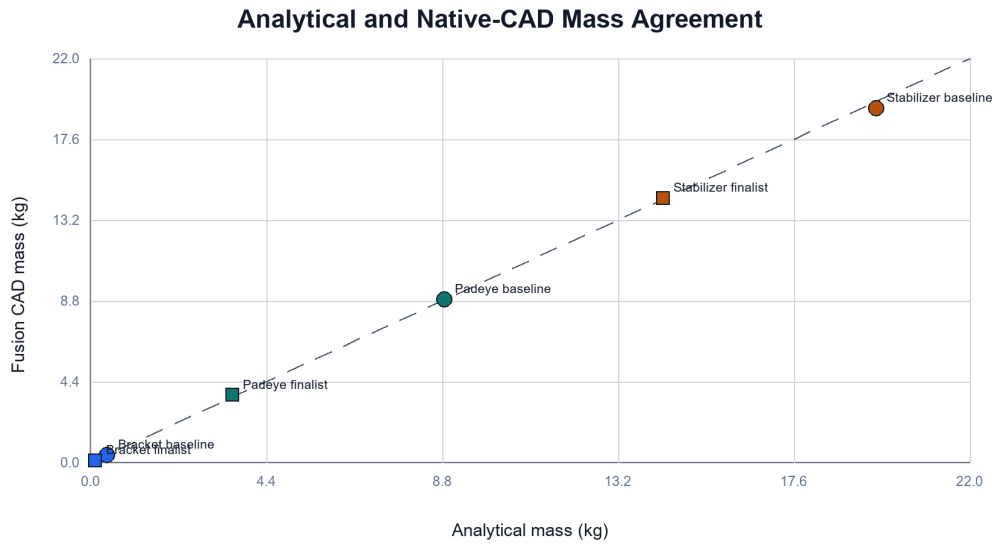
The stabilizer timeout is reported rather than hidden. The evidence supports successful artifact generation for all 21 vectors, but it does not support a claim that the complete outer integration test exited normally. The event also shows why long-running desktop automation should preserve intermediate artifacts and per-case records.

4.3.2 Analytical and Native-CAD Mass Comparison

Table 4.7 compares analytical and Fusion CAD mass for each baseline and selected finalist. The percentage difference is calculated relative to the analytical mass.

Table 4.7: Analytical and native-CAD mass comparison

Part	Design	Analytical (kg)	Fusion CAD (kg)	Difference
Bracket	Baseline	0.418125	0.422813	+1.12%
Bracket	Finalist	0.113396	0.114705	+1.15%
Padeye	Baseline	8.847625	8.897275	+0.56%
Padeye	Finalist	3.547772	3.682599	+3.80%
Stabilizer	Baseline	19.655593	19.297290	-1.82%
Stabilizer	Finalist	14.315361	14.400328	+0.59%

**Figure 4.3:** Analytical mass compared with Fusion native-CAD mass. The diagonal shows exact agreement.

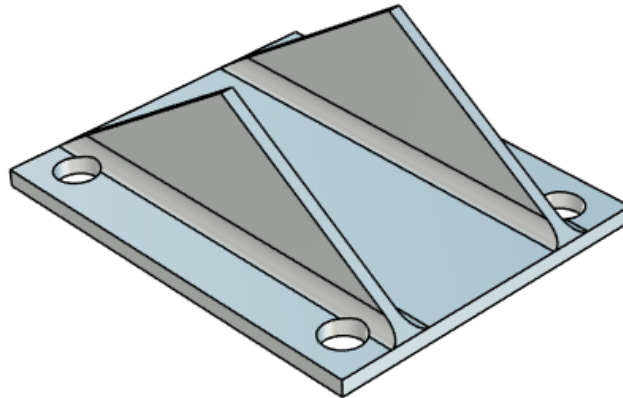
All six differences were within 3.8%. The agreement supports the broad volume approximations used for mass ranking. The padeye finalist showed the largest difference, which is consistent with greater geometric complexity around the lug crown, fillets, and gusset intersections.

The agreement does not validate stress. Mass depends primarily on volume and density. Structural response depends on the load path, section properties, local geometry, support idealization, contact, and mesh. The CAD response therefore remains correctly marked as partial.

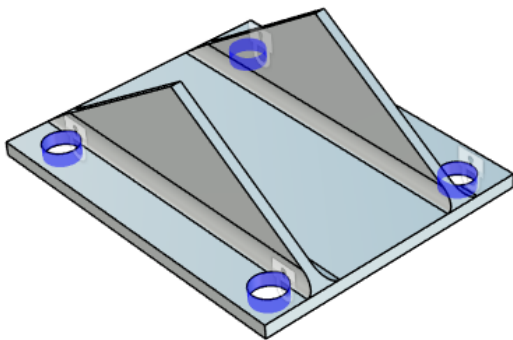
4.4 Bracket Static Stress Study Evidence

Two bracket Static Stress reports were available for review. Study 1 records the first exploratory solve, while Study 2 documents a revised setup with explicit images of the constrained and loaded faces. Both studies use the same bracket form and Aluminum

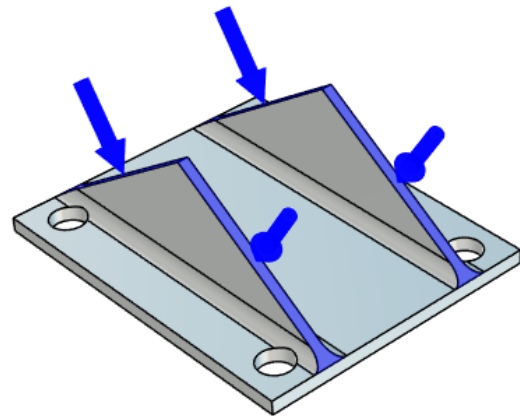
6061 material. Figure 4.4 shows the progression from the generated geometry to the more clearly documented boundary conditions.



(a) Study 1 bracket geometry



(b) Study 2 mounting-hole constraints



(c) Study 2 rib-face loading

Figure 4.4: Bracket geometry and boundary-condition evidence from the two Fusion studies. Study 2 makes the four constrained mounting holes and four loaded upper rib faces visible.

The support regions in Figure 4.4(b) correspond to the four cylindrical mounting-hole faces. This is a useful first idealization of a rigid bolt-down, although it does not represent bolt preload, clearance, bearing, or contact with the supporting structure. The arrows in Figure 4.4(c) confirm that the upper sloping faces of both ribs were selected for loading. The Study 2 data, however, record a total magnitude of 1.472 N with *force per entity* disabled. The intended total was 1472 N. The recorded force also contains both X and Z components, which makes direction another item that must be confirmed in the corrected study.

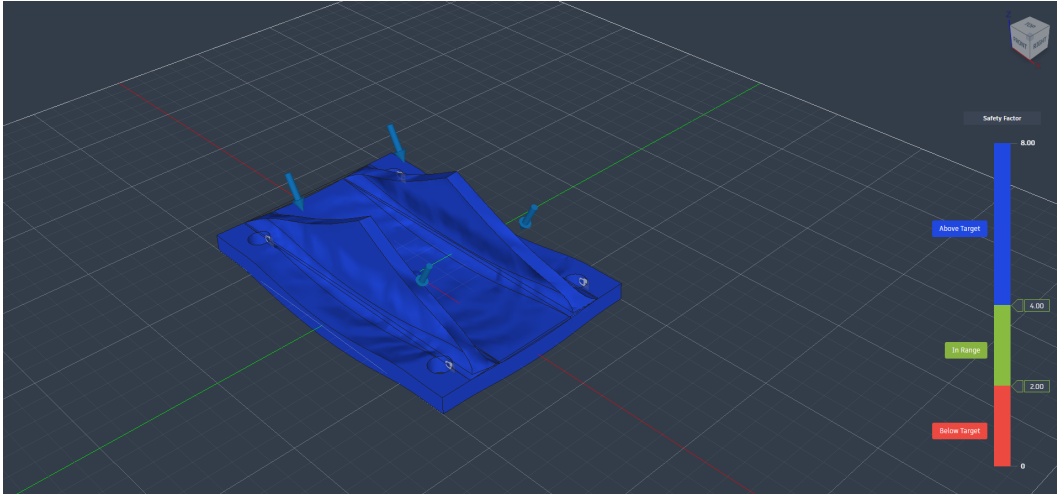
Table 4.8 summarizes the numerical evidence. Study 1 does not provide a complete applied-load record, but its reported 1.298 N support reaction and very small response show that it also operated at a newton-scale load. Study 2 improves traceability but

preserves the factor-of-one-thousand magnitude error.

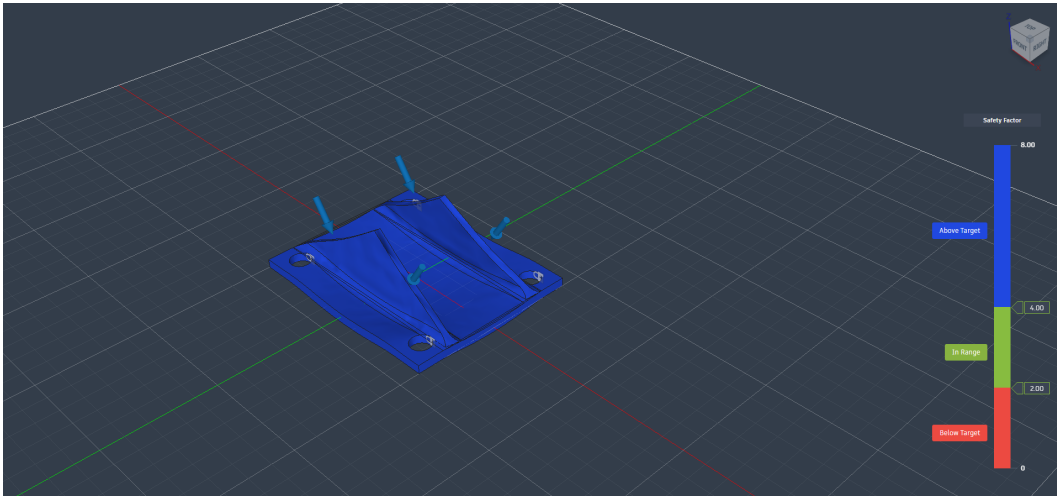
Table 4.8: Comparison of the two exploratory bracket studies

Recorded item	Study 1	Study 2
Load evidence	No complete applied-load table; reported support reaction magnitude 1.298 N	1.472 N total; force per entity disabled; $X = -0.815$ N and $Z = -1.226$ N
Mesh evidence	Parabolic elements; 10% average element size; no refinement step	Parabolic elements; 4781 nodes and 2355 elements; 10% average element size; no refinement step
Maximum von Mises stress	0.016 MPa	0.026 MPa
Minimum factor of safety	17 628	10 479
Maximum displacement	4.595×10^{-6} mm	9.223×10^{-6} mm
Engineering status	Rejected because the loading is incomplete and inconsistent with the 1.4715 kN design envelope	Rejected because 1.472 N was applied instead of 1472 N; direction and reaction balance also require confirmation

The factor-of-safety plots in Figure 4.5 are almost uniformly blue because the display is capped at 8 while the reported minimum factors of safety exceed ten thousand. The images are therefore evidence of an unrealistically lightly loaded model, not evidence of a highly efficient bracket. They also illustrate why a colour plot must be read together with its legend and input table.



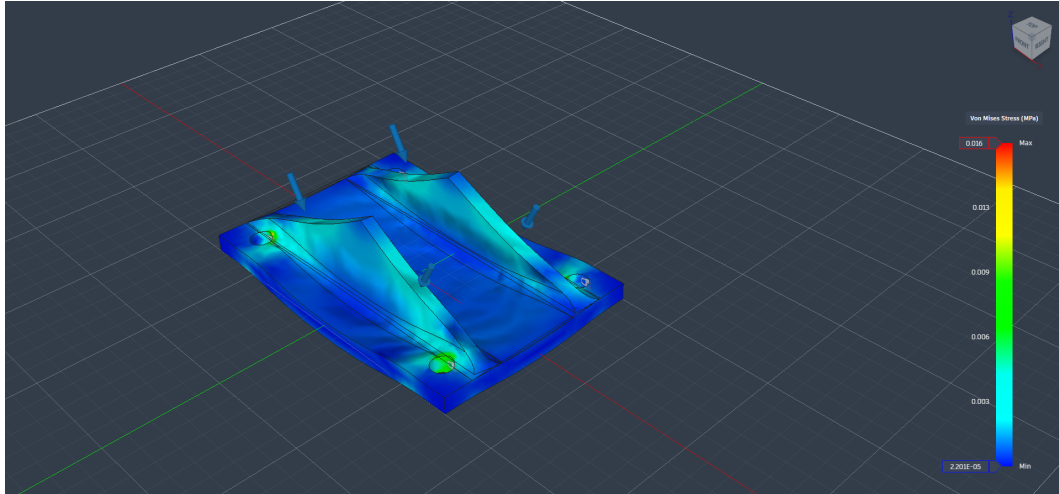
(a) Study 1 factor-of-safety plot



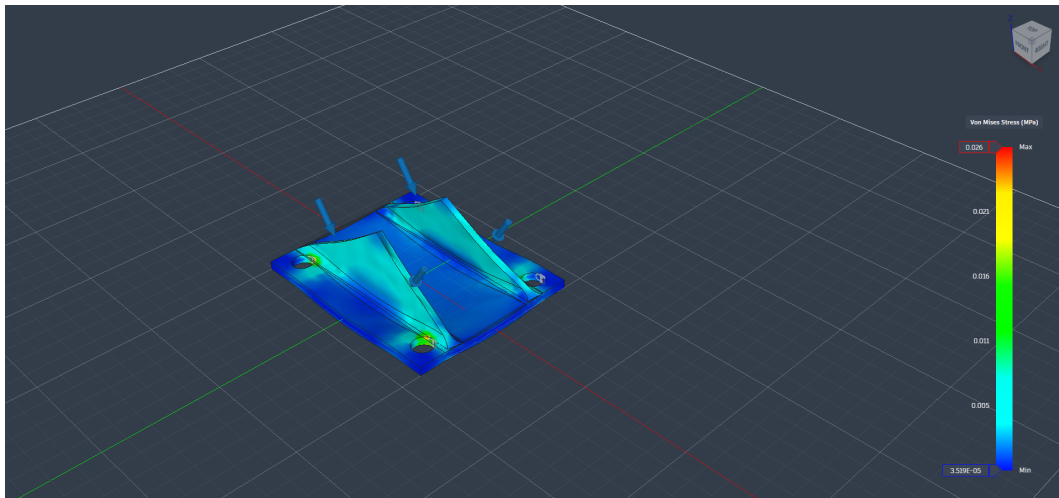
(b) Study 2 factor-of-safety plot

Figure 4.5: Factor-of-safety plots from Studies 1 and 2. The common display cap of 8 masks the much larger reported values produced by the incorrect newton-scale loading.

The von Mises plots in Figure 4.6 show higher stress around the mounting-hole edges, rib roots, and load-introduction regions. These locations are consistent with the expected load path and identify where local mesh refinement and more realistic bolt/contact modelling would be required. The maxima of 0.016 MPa and 0.026 MPa cannot be used for acceptance because neither study represents the design load.



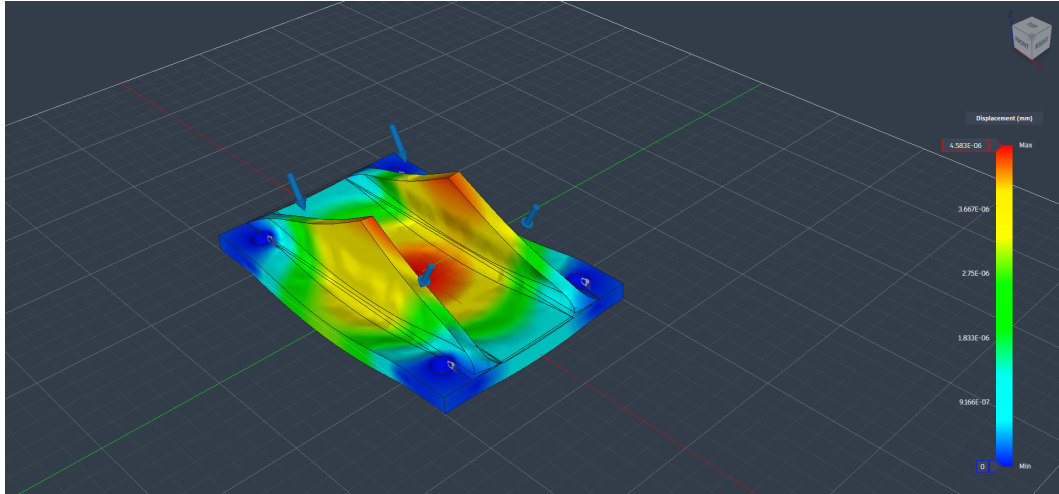
(a) Study 1 von Mises stress



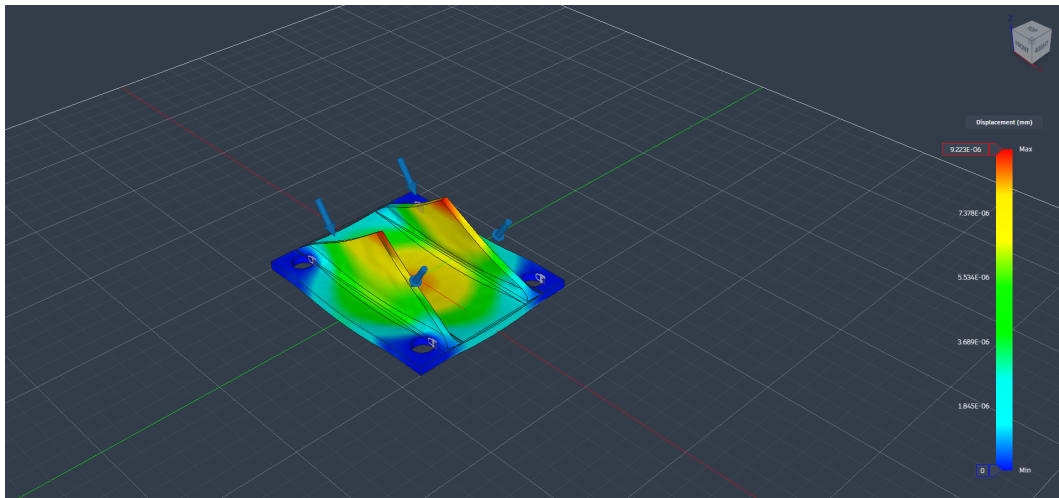
(b) Study 2 von Mises stress

Figure 4.6: Von Mises stress distributions from the two exploratory studies. Only the qualitative locations are retained as useful evidence.

Figure 4.7 shows minimum displacement near the constrained mounting holes and larger movement across the unconstrained plate and rib regions. This is qualitatively consistent with the selected supports. The two maximum values remain several orders of magnitude below the analytical screening displacement because the applied study loads were several orders of magnitude too small. No calibration conclusion can therefore be drawn from the apparent difference between the analytical and solver responses.



(a) Study 1 total displacement



(b) Study 2 total displacement

Figure 4.7: Total-displacement distributions from Studies 1 and 2. The spatial patterns are informative, but the magnitudes are invalid for the project load.

The bracket studies therefore remain valuable as audit evidence. They confirm that the native geometry reached the Simulation workspace, show the intended support and load regions, and reveal plausible qualitative response locations. They also demonstrate that solver completion is not an acceptance criterion. A corrected 1472 N study must verify material data, direction, reaction balance, contact assumptions, local mesh refinement, and convergence before stress, factor of safety, or displacement is compared with the analytical finalist.

4.5 Discussion

4.5.1 Achievement of the Project Objectives

Table 4.9: Achievement of the specific objectives

No.	Objective outcome	Status
1	Unit-aware parameter spaces, materials, loads, constraints, screening models, and boundary roles were defined for all three components.	Achieved for analytical screening and CAD preparation.
2	Each part completed 60 seeded LHS evaluations and three bounded Nelder–Mead starts; failed evaluations remained controlled.	Achieved.
3	Native Fusion geometry, physical properties, exchange artifacts, and semantic boundary evidence were generated.	Achieved for serialized CAD requests.
4	Baseline plus 20-vector CAD sweeps and baseline/finalist validation sets were completed.	Achieved, with the stabilizer outer-process timeout disclosed.
5	Lower-mass analytical finalists were selected and their mass estimates were compared with native CAD.	Achieved as screening evidence; all mass differences were within 3.8%.
6	The bracket proof of concept reached manual Static Stress review, and the remaining evidence required for design acceptance was identified.	Partly achieved; both exploratory solver studies were rejected and a corrected 1472 N study remains required.

4.5.2 Limitations of the Results

The analytical models are intentionally first order. They use simplified load paths, nominal section properties, and approximate concentration factors. The reported constraints include yield-based factor of safety and displacement only.

The search used a finite sample and three local starts. A different seed, start, penalty, bound, or model can produce a different candidate. Several finalists lie near lower bounds, so the selected bounds strongly influence the result.

The CAD sweeps verify sampled geometry, not the full continuous design space. They also do not check manufacturing drawings, tolerances, weld access, or assembly.

The manual FEA evidence is incomplete. The available bracket report has an incorrect load and cannot validate stress or displacement. Padeye and stabilizer solver campaigns remain future work. No physical tests were performed.

4.5.3 Engineering Interpretation

The results support the central workflow claim: inexpensive analytical evaluation can reduce a large design set to a small group of traceable candidates, and native CAD can check geometry and mass before manual FEA. They do not support certification or fabrication.

The most important result is not the largest percentage reduction. It is the separation of claims. The system recorded analytical predictions, returned partial CAD responses without solver values, disclosed the stabilizer timeout, and rejected a completed Fusion report after finding the wrong load. This behavior is consistent with an auditable engineering workflow.

Chapter 5

Project Workplan

This chapter presents the project schedule from September 2026 to March 2027. The workplan groups the study into research and engineering definition, implementation, verification, and reporting activities. Detailed technical methods and findings are presented in Chapters 3 and 4, respectively.

5.1 Project Schedule

The workplan spans seven months. September and October establish the research basis, engineering definitions, and bracket proof of concept. November and December develop the design-space search, result records, and multi-part architecture. January and February focus on native CAD generation, verification, validation preparation, and analysis of the results. March is reserved for final reporting, demonstration, corrections, and submission.

The Fusion proof of concept provides the principal technical decision point. Because the public API does not provide dependable automated control of the Static Stress workspace, the workflow retains analytical optimization and native CAD generation while treating solver assessment as a controlled manual activity. Figure 5.1 summarizes the sequence. Documentation and report writing overlap the engineering work because assumptions, decisions, and evidence are recorded throughout the project.

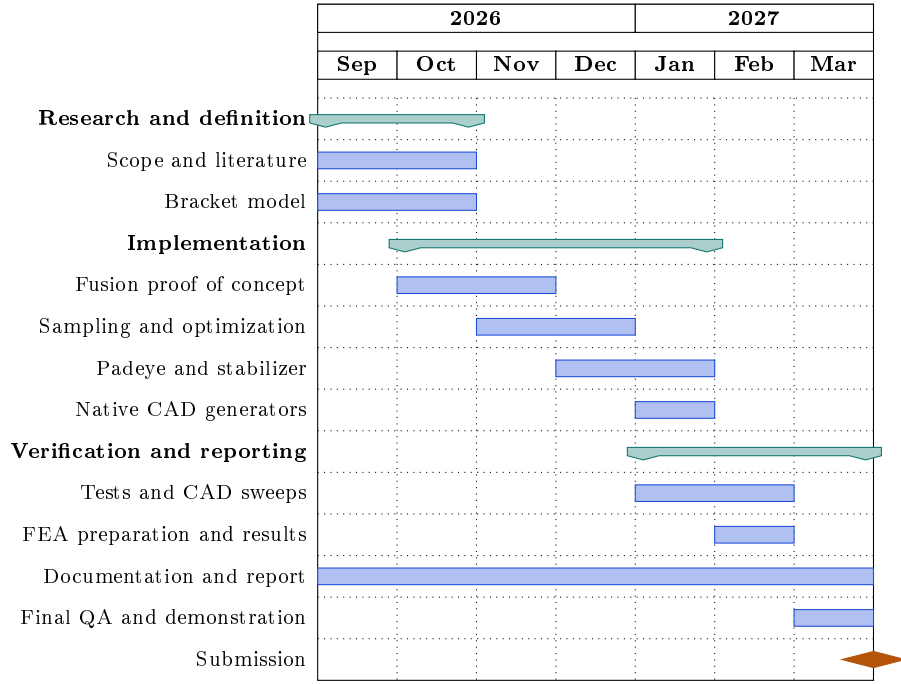


Figure 5.1: Project workplan for September 2026 to March 2027

5.2 Workplan Review

The schedule allows analytical development and CAD verification to proceed before the final simulation review. This sequencing limits the number of candidates that require higher-fidelity assessment. It also provides time in March for resolving report corrections and presenting the evidence generated by the project. Where Fusion-dependent work is delayed, the analytical models, recorded CAD requests, and validation packages preserve progress and allow the affected activity to resume without repeating the full design-space search.

Chapter 6

Project Budget

This chapter presents the direct-cash budget and the in-kind resources used by the project. The work is planned around existing computing facilities, educational software access, open-source tools, and digital submission. Consequently, the direct-cash budget is zero, although the project still depends on material institutional and personal resources.

6.1 Direct-Cash Budget

No equipment, commercial software, cloud computation, data, printing, travel, fabrication, or testing service is purchased from project funds. Table 6.1 summarizes the resulting budget.

Table 6.1: Direct-cash project budget

Category	Basis for zero direct expenditure	Budget (KES)
Development hardware	Existing personal or university computers are used. No computer or upgrade is purchased for the project.	0
CAD and simulation software	Autodesk educational access is used. No commercial licence is purchased from project funds.	0
Python and report software	Python, project libraries, Git, and the LaTeX toolchain are available without direct project charges.	0
Cloud computing and storage	Computation and working data remain local or use existing remote access. No paid cloud service is purchased.	0
Internet and electricity	Existing personal or university services support the work. No separate project subscription or utility purchase is budgeted.	0
Printing, travel, and communication	The report is prepared for digital submission. No project-specific printing, travel, or communication expense is charged.	0
Prototype fabrication and testing	No physical component or paid laboratory test is included in the project scope.	0
Contingency	The contingency consists of schedule margin and scope control rather than a cash allocation.	0
Total direct-cash budget		0

6.2 In-Kind Resources

A zero direct-cash budget does not imply that the project uses no resources. Existing computers, Autodesk educational access, internet, electricity, supervision, and student labour are in-kind inputs. Table 6.2 records these resources without assigning unsupported monetary values.

Table 6.2: In-kind resources used but not monetized

Resource	Provision basis	Project use
Computers	Existing personal or university access	Python development, testing, Fusion CAD, simulation, report compilation, and demonstration.
Autodesk Fusion	Educational entitlement	Native CAD generation, physical-property evaluation, STEP export, and Static Stress preparation.
Open-source software	Public software licences	Optimization, testing, data recording, visualization, version control, and report production.
Internet and electricity	Existing personal or institutional access	Research, reference consultation, source synchronization, and software operation.
Student labour and supervision	Academic project activity	Engineering definition, implementation, review, testing, analysis, documentation, and presentation preparation.

6.3 Budget Control

The project architecture supports the zero direct-cash budget by using analytical equations for the broad design-space search and reserving manual FEA review for a small number of candidates. The in-application Fusion API and local data exchange avoid paid cloud automation, while open-source Python tools and digital reporting avoid additional software and printing costs.

Prototype fabrication and paid testing remain outside the present academic scope; they are not treated as unnecessary. Progression towards a service-ready design requires a separate budget for material certification, fabrication, inspection, instrumentation, proof loading, calibrated FEA, and any required hydrodynamic testing.

Chapter 7

Conclusions and Recommendations

This chapter presents the conclusions supported by the analytical, CAD, and bracket-study evidence. It also states the limitations of the work and the validation required before any candidate can progress toward manufacture or service.

7.1 Conclusions

The study developed and evaluated an auditable, resource-conscious workflow for the parameterized mass optimization of three marine structural component archetypes. The same overall process represented a bracket, padeye, and stabilizer while allowing each component to retain its own variables, material, load, analytical formulation, CAD generator, and boundary roles. This supports the use of a shared part-aware architecture within the reported scope.

Seeded Latin-hypercube exploration and three bounded Nelder–Mead starts identified lower-mass candidates that satisfied the implemented analytical factor-of-safety and displacement criteria. The selected screening reductions were 72.9% for the bracket, 59.9% for the padeye, and 27.2% for the stabilizer relative to their configured baselines. Multiple starts improved the best feasible mass for every part. These values describe the best feasible designs found within the declared parameterized searches; they do not prove global optimality or service-ready weight savings.

Native Fusion generation remained usable across the seeded CAD campaigns. The bracket and padeye completed all 21 baseline-plus-sample cases. The stabilizer created all 21 artifact sets, although the outer process timed out while reading the final response. Analytical and native-CAD mass differed by no more than 3.8% for the six baseline and finalist comparisons. This agreement supports the volume models for early mass ranking, but it does not validate stress or displacement.

The two bracket Static Stress studies provide the clearest lesson about evidence quality. Their images identify the geometry, mounting-hole supports, rib-face loading, and qualitative response regions. Study 2 also records the mesh and boundary setup

more clearly than Study 1. Nevertheless, Study 2 used 1.472 N instead of the intended 1472 N, while Study 1 did not provide a complete traceable load definition and reported only a 1.298 N support reaction. The numerical stress, factor of safety, and displacement values from both studies were therefore rejected.

The principal outcome is a traceable method for reducing a large design set, checking native geometry and mass, and preparing a small candidate set for disciplined validation. The study does not establish structural qualification, fabrication readiness, or service safety.

7.2 Achievement of the Objectives

The aim was achieved at the level of workflow development, analytical screening, native-CAD verification, and validation preparation. Unit-aware engineering definitions and screening models were completed for all three parts; the sampling and multi-start searches were executed; invalid evaluations were contained; native geometry, physical properties, exchange artifacts, and boundary evidence were produced; and analytical-to-CAD mass agreement was assessed. Table 4.9 records the status of each specific objective.

The aim was only partly achieved at the structural-validation level. The bracket reached manual FEA review, but the two exploratory studies did not use a sufficiently traceable and correct design load. Corrected, mesh-converged FEA has not yet been completed for the selected bracket, padeye, or stabilizer candidates.

7.3 Limitations

The analytical models are first-order representations of selected load paths. They omit detailed fastener and weld behaviour, nonlinear contact, three-dimensional local stress, fatigue, buckling, corrosion, vibration, impact, and other component-specific service effects. The stabilizer load is a prescribed structural envelope rather than a pressure field derived from CFD, model tests, or measurements.

The search uses a finite 60-point sample, three local starts, fixed bounds, and a penalty objective. A different seed, model, bound, or constraint strategy could produce a different finalist. Several selected parameters lie near their lower bounds, which indicates that manufacturing limits and omitted failure modes can materially affect the result.

The CAD sweeps cover seeded design vectors rather than every point in the continuous space. They verify geometry, mass, artifacts, and boundary identification, but do not verify drawings, tolerances, assembly, weld access, or manufacturability. No acceptable finalist FEA or physical test result is available.

7.4 Recommendations

1. Recreate the bracket study with a verified total load of 1472 N, confirm its coordinate direction, keep force per entity disabled, and demonstrate reaction balance. Compare the baseline, lightest finalist, and conservative backup candidates using at least three systematically refined meshes.
2. Replace the rigid hole constraint with more realistic bolt, bearing, contact, and preload representations when these effects influence the bracket decision. Assess local plate behaviour, rib-root and weld stress, fatigue, corrosion allowance, fabrication tolerance, and supporting structure.
3. Validate the padeye under the governing lifting standard and load combinations. Begin with a distributed bearing model and progress to a separate pin with clearance and nonlinear contact where required. Check net section, bearing, shear-out, lug root, gussets, welds, fatigue, inspection, and proof-load requirements.
4. Derive stabilizer pressure cases from an accepted hydrodynamic method, CFD, model tests, or measurements. Include positive and negative loading, torsion, root and spar stresses, skin-panel buckling, fatigue, corrosion, actuator interaction, vibration, and tip displacement.
5. Compare corrected FEA responses with the analytical predictions at identical candidate geometries and loads. Any calibration should use separate fitting and checking candidates and remain limited to the supported parameter range.
6. Treat the lightest analytical row as one candidate rather than the automatic design choice. Material certification, applicable codes, uncertainty, fabrication, inspection, physical testing where required, and qualified engineering review must precede acceptance.

7.5 Future Work

Future development can introduce a formal multi-fidelity strategy in which selected FEA results correct analytical bias or support a surrogate model with explicit uncertainty. Reliability-based optimization can then represent variability in loads, material properties, fabrication dimensions, corrosion, and modelling error.

The search can also be extended from single-objective mass minimization to a multi-objective formulation that includes fabrication cost, fatigue damage, displacement, or embodied carbon. Manufacturing-aware constraints should represent minimum gauges, available stock sizes, tool and inspection access, weld size, bend radii, and tolerance sensitivity.

For the individual components, the bracket can be extended to equipment vibration and foundation flexibility, the padeye to lifting-operation load angles and proof testing, and the stabilizer to coupled hydrodynamic and structural loading. Each extension should preserve the distinction between a screening prediction, a CAD property, a solver result, and a validated design.

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Appendix A

Supplementary Data

This appendix contains detailed candidate dimensions and supplementary result plots that support the discussion in the main report. Reproduction commands, software interfaces, and operating instructions remain in the project documentation and are not repeated here.

A.1 Selected Finalist Parameters

Table A.1: Selected bracket finalist parameters

Parameter	Value (mm)
Baseplate length	100.0328
Baseplate width	80.9380
Baseplate thickness	4.0429
Rib height	33.2576
Rib thickness	3.0000
Fillet radius	3.5308

Table A.2: Selected padeye finalist parameters

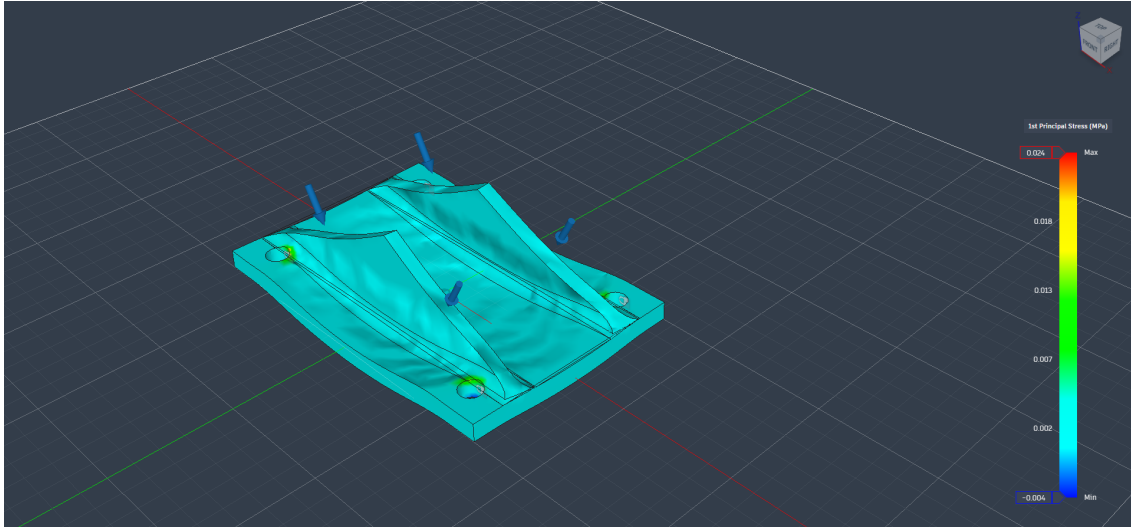
Parameter	Value (mm)
Base length	220.3878
Base width	150.4654
Base thickness	8.0484
Lug height	141.7256
Lug thickness	12.1613
Neck width	80.0000
Gusset height	70.0000
Gusset thickness	6.2101
Fillet radius	25.0000

Table A.3: Selected stabilizer finalist parameters

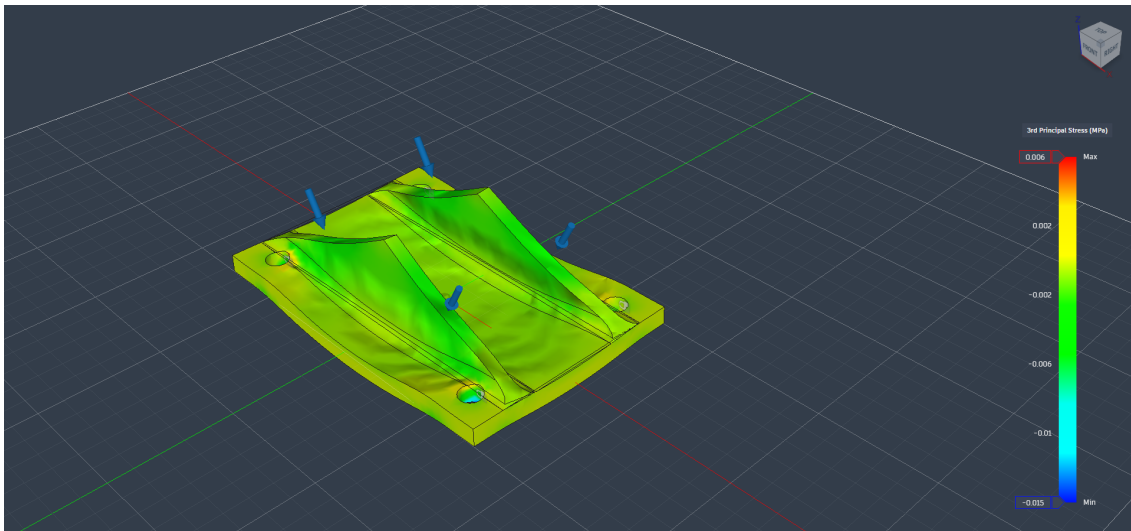
Parameter	Unit	Value
Skin thickness	mm	4.0106
Front spar position	% chord	25.0000
Rear spar position	% chord	72.0000
Front spar thickness	mm	7.8522
Rear spar thickness	mm	6.5297
Rib thickness	mm	5.6971
Root insert length	mm	221.4762
Root insert thickness	mm	13.6624
Root fillet radius	mm	18.3273

A.2 Supplementary Bracket Stress Plots

Study 1 includes first- and third-principal stress plots in addition to the von Mises result discussed in Chapter 4. Figure A.1 shows tensile and compressive regions around the mounting holes, rib roots, and load-introduction surfaces. The distributions support the identification of locations for later mesh refinement, but their magnitudes are not valid for the 1.4715 kN design load.



(a) First-principal stress



(b) Third-principal stress

Figure A.1: Study 1 principal-stress plots. The distributions are retained only as qualitative evidence because the study load is not acceptable.

The Study 2 third-principal plot in Figure A.2 again places the largest compressive response near mounting-hole and rib-transition regions. This agreement in location is useful for planning a corrected model, but it does not make either exploratory result quantitatively valid.

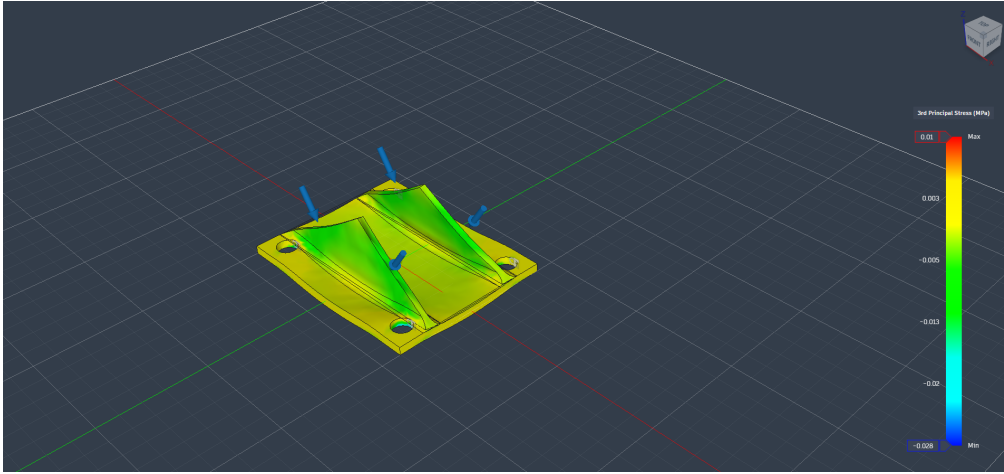


Figure A.2: Study 2 third-principal stress distribution. The magnitude is rejected because the applied load was 1.472 N instead of 1472 N.

A.3 Stabilizer Pressure-Band Distribution

The prescribed stabilizer resultant was distributed over four spanwise bands using the relative weights

$$\mathbf{w} = [1.00, 0.93, 0.75, 0.40]. \quad (\text{A.1})$$

The normalized fractions and corresponding forces are given in Table A.4. Pressure on each band is obtained by dividing its assigned force by the corresponding CAD surface area.

Table A.4: Prescribed stabilizer pressure-band forces

Band	Normalized fraction	Force (N)
1	0.324675	3833.8
2	0.301948	3565.4
3	0.243506	2875.1
4	0.129870	1533.8
Total	1.000000	11 808.0

These loads are declared structural inputs, not CFD predictions. A later hydrodynamic study must establish the operational pressure field and relevant positive, negative, and torsional load cases.